

THE JUDGE PROTOCOL

TJP

A Framework for Human Oversight of Artificial Intelligence

Working Document — Version 0.95 | July 2026 | Author: Alexander Hofmann

⚠ WORKING DRAFT — This document is a working draft for discussion and development purposes only. Content, architecture and specifications are subject to revision.

© 2026 Alexander Hofmann. All rights reserved.

This document is released under Creative Commons Attribution 4.0 International (CC BY 4.0). You are free to share and adapt this material for any purpose, provided appropriate credit is given to Alexander Hofmann as the original author, a link to the licence is provided, and any changes are indicated. Full licence text:

<https://creativecommons.org/licenses/by/4.0/>

DOI: 10.5281/zenodo.20425548

Correspondence: Alexander Hofmann | Stockholm, Sweden | judgeprotocol@pm.me

Version History

v0.80 | February 2026 | Initial working draft

v0.85 | March 2026 | First external distribution

v0.91 | April 2026 | HACK architecture expanded; physical domain added; governance structure completed

v0.92 | May 2026 | Copyright, licence and DOI added; author name standardised to Alexander Hofmann

v0.93 | May 2026 | Geneva Lineage revised; Foundation registered in Sweden; PACK defined; HACK definition expanded; open standard framing applied

v0.94 | July 2026 | New "What HACK Actually Is" subsection added to Section 4 (design philosophy surfaced from Appendix B); new Redirect Gate subsection added to Section 7 and detailed specification added to Appendix C (mechanism, National Response Registry, extensibility statement); five use-case rule-set placeholders added, nested under their matching Appendix F domain: F1.4 and F6.1 (AI Misuse for Private Gains), F2.3 and F5.1 (Human Life Impact), F6.2 (Cyber Misuse); open-problem notes added for data protection (Appendix C) and hardware partner dependency (Appendix B)

v0.95 | July 2026 | Removed and cleaned §7.1–§7.11 mapping, adding TJP for The Judge Protocol; corrected title-page DOI (was incorrectly showing the Executive Summary's DOI — now shows the Main Protocol's own concept DOI, 10.5281/zenodo.20425548)

Table of Contents

Foreword.....	7
1. Problem Statement.....	8
The Governance Gap.....	8
Seven Documented Failure Domains.....	9
The Governance Vacuum.....	10
2. The Judge Protocol — Core Concept	11
Architectural Foundation — The TCP/IP Principle.....	12
3. The Judge Protocol — Rules.....	14
Rule 1: The Human Override Requirement.....	14
Rule 2: The Escalation Ceiling	14
Rule 3: The Transparency Declaration	14
Rule 4: The Accountability Chain	14
Rule 5: The Reversibility Principle.....	14
Rule 6: The Evolution Clause.....	15
Rule 7: The Universal Membership Requirement.....	15
4. The Decision Gate Mechanism.....	15
What HACK Actually Is	15
Decision Gate Classification Tiers	16
The Decision Gate Workflow	16
Emergency Change Process	17
Audit Trail Requirements	18
5. Proposed Governance Structure.....	19
Voting Rights and Decision Making.....	19
Relationship with Existing International Bodies	20
Headquarters and Neutrality.....	20
6. National Implementation Framework	21
The Representative Lead	21
The Support Function.....	22
The Global NAIA Network	22
Minimum Capability Standards.....	23
Private Sector Obligations.....	23
7. Global Emergency Dashboard.....	24
Redirect Gate	25
8. Funding Framework	25
The Core Challenge	25
Reference Model — CERN	26

Industry Contribution Models — Initial Proposals.....	26
Independence Safeguards	27
Next Steps on Funding	27
9. Physical Domain — Robots, Humanoids and Autonomous Systems.....	29
Physical Domain Robot Categories	29
The Physical ACK Concept.....	30
The Latency Challenge	30
Autonomous Weapons — The Hard Line.....	31
The Moral Agency Question	32
Category P4 — Private Service: Special Considerations	33
Who Holds Authorization Authority in a Private Home?.....	33
Vulnerable User Protections	33
Privacy — The Home Environment Challenge	34
The Witness Obligation Question	35
Liability Framework for P4	35
10. Why Now.....	36
An Intellectual Lineage — From CERN to the Judge Protocol.....	36
11. Next Steps — Open Questions for Development	37
References	39
International Governance Frameworks	39
Technical Standards and Architecture	39
AI Risk and Governance Research.....	40
Institutional References and Policy References.....	40
Appendix A — RASI Matrix: Roles & Responsibilities	41
Appendix B — Technical Infrastructure: Conceptual Overview	43
Core Infrastructure Principles	43
Global Infrastructure Layer (IAIA operated)	43
National Infrastructure Layer (NAIA operated)	44
Decision Gate Technology Requirements.....	45
Cybersecurity Requirements.....	45
Phased Implementation.....	46
Silicon-Level Enforcement — The Hardware ACK.....	46
Photonic Chips — The Direction of Travel	48
Open Problem — Hardware Partner Dependency	48
Appendix C — Global Emergency Dashboard: Detailed Specification.....	49
Design Principles	49
Dashboard Components	50
Alert Level Transition Rules	51

Notification Protocol.....	51
Public Transparency	52
Redirect Gate: Detailed Specification	52
National Response Registry (NRR).....	52
Open Problem — Data Protection.....	52
Appendix D — Network, Hosting and Data Infrastructure	54
Why Standard Commercial Infrastructure is Insufficient	54
Infrastructure Design Principles.....	55
Global Layer Infrastructure Requirements	56
National Layer Infrastructure Requirements	56
The CERN Infrastructure Model.....	58
Phased Infrastructure Deployment	58
Appendix E — Glossary of Terms	59
Appendix F — Domain Specific Rules	65
F1 — Financial Markets.....	65
F1.1 — Financial Market Decision Gate Classification	65
F1.2 — Financial Market Specific Rules	66
F1.3 — Open Issues Requiring Dedicated Working Paper	67
F1.4 — Use Case: AI Misuse for Private Gains	68
F2 — Healthcare	68
F2.1 — Healthcare Decision Gate Classification	68
F2.2 — Healthcare Specific Rules.....	69
F2.3 — Use Case: Human Life Impact	69
F3 — Military and Defense	70
F4 — Critical Infrastructure.....	70
F5 — Governmental and Public Administration	70
F5.1 — Use Case: Human Life Impact	71
F6 — Judicial	71
F6.1 — Use Case: AI Misuse for Private Gains	71
F6.2 — Use Case: Cyber Misuse.....	71
F7 — Transport	71
F8 — Education	71
F9 — Intelligence and Surveillance	72
F10 — Private Domain (P4).....	72
F11 — AI-to-AI Communication	72
Appendix G — AI-to-AI Communication and Transition Arrangements	73
G1 — AI-to-AI Communication Governance	73

G1.1 — The AI-to-AI Communication Problem	73
G1.2 — AI-to-AI Communication Rules.....	74
G2 — Transition Arrangements	75
G2.1 — Transition Principles.....	75
G2.2 — Transition Timeline by Tier	75
G2.3 — Transition Support Programme	76
G2.4 — Non-Member State Provisions.....	77

Foreword

Alexander Hofmann — May 2026

This document began with a simple observation, not a grand theory. As an Enterprise Architect with over thirty years in the IT industry — across finance, technology and infrastructure — I have governed consequential decisions through the same basic mechanism: a human reviews, a human authorizes, a human is accountable. The Change Request. The approval workflow. The sign-off chain. These are not sophisticated inventions. They are the accumulated common sense of organisations that learned, often the hard way, that consequential decisions need a human in the loop.

The insight that produced the Judge Protocol was not complicated: the governance frameworks we need already exist. TCP/IP tells us no packet is delivered without an acknowledgement. The Change Request process tells us no consequential system change proceeds without human review. The IAEA tells us binding international oversight of dangerous technology is achievable even among states that do not trust each other. None of this needed to be invented. It needed to be applied.

What follows is my attempt to do that — to take the governance principles that work in every well-run institution in the world and construct from them a universal framework for human oversight of AI. The framework is called the Judge Protocol, named for the mandatory human judgment layer that must sit between what AI recommends and what AI executes.

I wrote this document in collaboration with Claude, an AI model developed by Anthropic. That is worth acknowledging directly, because it is part of the story. Claude did not author the Judge Protocol — the ideas, the decisions and the positions are mine. But the collaboration was itself a demonstration of the principle the protocol is designed to protect: human authority, AI capability, in the right order.

This is a working document — version 0.95 of a framework that will require expert review, legal analysis and political negotiation before it could become the binding international instrument it is designed to be. The window to establish AI governance norms before they are urgently needed is closing. This is my contribution to using it.

Alexander Hofmann

Stockholm, May 2026

1. Problem Statement

The Governance Gap

Artificial Intelligence is no longer a laboratory concept. It is embedded in financial markets, military planning, healthcare systems, critical infrastructure, private homes and physical robots operating in the world alongside humans. The pace of AI integration has dramatically and consistently outpaced the development of governance frameworks capable of managing its risks.

The central danger is not that AI will malfunction — it is that AI will function exactly as designed, but without adequate human judgment in the decision loop. Automated systems can escalate conflicts, trigger market crashes, execute irreversible physical actions, make life and death clinical decisions, and propagate errors at speeds that make meaningful human intervention impossible after the fact.

We have already largely ceded control of financial markets, military targeting support and critical infrastructure management to AI systems. We are in the process of ceding control of healthcare decisions, physical environments and personal care to humanoid robots and service AI. We are doing this without a binding global governance framework that ensures humans retain ultimate decision-making authority over consequential actions.

This is not a future risk. It is the present reality.

Seven Documented Failure Domains

The following failure domains have been identified, documented and in several cases are actively occurring. Each represents an area where ungoverned AI action poses risks of irreversible, large-scale harm:

- **Military escalation** — King's College London war game studies showed that leading AI models recommended nuclear strikes in 95% of simulated crisis scenarios, treating tactical nuclear weapons as routine escalation options rather than moral thresholds. No model ever chose accommodation or de-escalation — when losing, they escalated or died trying. As Professor Kenneth Payne, who led the study, concluded: understanding how frontier models do and do not imitate human strategic logic is essential preparation for a world in which AI increasingly shapes strategic outcomes. AI systems are currently embedded in active military planning and target identification in the ongoing 2026 Gulf conflict, without a universally agreed governance framework.
- **Financial system instability** — AI systems execute the majority of global financial trades, price the majority of derivatives and manage risk models across interconnected global markets. Ungoverned AI action can trigger systemic cascades faster than any human regulator can intervene. The 2010 Flash Crash was an early demonstration. AI systems are now orders of magnitude faster and more interconnected.
- **Healthcare decision autonomy** — AI systems are making or materially influencing diagnostic and treatment decisions affecting millions of patients globally. Without governance frameworks ensuring human clinical authority, patients face the risk of consequential medical decisions made without meaningful human judgment.
- **Physical domain** — autonomous robots — humanoids, autonomous vehicles, industrial robots and military autonomous weapons systems are operating in physical environments shared with humans. Physical AI actions are irreversible by default. A software AI that acts without authorization can potentially be corrected. A robot that harms a person cannot un-harm them.
- **Private domain vulnerability** — AI systems are entering private homes and personal spaces as care robots, companion humanoids and personal assistants. They interact with the most vulnerable members of society — children, elderly persons, individuals with disabilities — without adequate governance frameworks protecting against harm, dependency or exploitation.
- **AI-to-AI communication loops** — AI systems are increasingly communicating with, influencing and responding to each other in loops that no human designed or authorized. These loops can produce emergent behaviors, cascade amplifications and authorization laundering — outcomes that exceed what any individual system's human principals intended or approved.

- Infrastructure sovereignty — the global AI systems that govern critical infrastructure, financial markets and military operations increasingly run on infrastructure subject to the jurisdiction of a small number of powerful states and commercial entities. This creates systemic dependency and sovereignty risks that no current governance framework addresses.

The Governance Vacuum

Existing governance frameworks — national AI regulations, financial market rules, medical device regulations, arms control treaties — address specific domains within specific jurisdictions. None of them address the cross-domain, cross-border, cross-sector nature of AI risk. None of them establish a binding global framework for human decision-making authority. None of them govern the physical domain of humanoids and autonomous systems. None of them address AI-to-AI communication. None of them require the infrastructure sovereignty protections that the global deployment of consequential AI demands.

The result is a governance vacuum — a world in which the most powerful and consequential technology in human history operates without a universal framework for human oversight. This is the problem the Judge Protocol is designed to solve.

The absence of a binding global framework governing AI decision-making authority is not merely a policy gap — it is an existential governance failure that grows more dangerous with every passing month that AI capability advances faster than the frameworks designed to govern it.

2. The Judge Protocol (TJP) — Core Concept

The Judge Protocol is a proposed global governance framework that establishes a mandatory human decision layer for all consequential AI actions. It does not seek to limit AI capability or innovation — it seeks to ensure that humans retain ultimate decision-making authority where the consequences are irreversible, large-scale or affect fundamental rights.

The core principle is simple:

"AI may analyze, recommend and simulate — but a human must authorize."

This principle applies universally — across military, financial, medical, judicial and governmental domains — wherever AI systems have the capacity to initiate or execute consequential actions.

The protocol is built on three foundational design principles that address the most common political and practical objections to global AI governance:

Foundational Principle	What it means in practice
Sovereignty by Design	National data remains under national jurisdiction at all times. The global layer aggregates only metadata and compliance signals — never raw content. No government is required to share sensitive AI system data with the IAIA or any other state. This is not a compromise — it is a core architectural requirement of the protocol.
Human Authority — Always	AI may analyze, recommend and simulate. But a human must authorize every consequential action. This applies regardless of the speed, complexity or domain of the AI system. No exception exists for operational urgency — the Emergency Change Process provides the time-critical pathway while preserving human authority.
Evolution by Design	The protocol is a living framework, not a static treaty. It is designed to evolve as AI capability evolves — with a mandatory review cycle and a defined governance process for updates. The protocol does not attempt to anticipate every future AI development; it establishes the governance architecture within which future rules can be made.

Architectural Foundation — The TCP/IP Principle

The Judge Protocol is not a novel governance invention — it is the application of a proven technical architecture to a new domain. Its decision mechanism is deliberately modeled on two of the most successful and universally adopted standards in the history of computing: the TCP/IP protocol stack and the network firewall.

TCP/IP — the protocol that underpins the entire internet — is built on a single foundational principle:

No packet is considered delivered until an explicit acknowledgement (ACK) is received.

The Judge Protocol applies this same principle to AI governance:

No AI action is considered authorized until an explicit human acknowledgement is received.

TCP/IP alone handles communication and confirmation — but it does not handle permanent refusal. That is the role of the firewall. A firewall applies rule-based hard stops to packets that match defined criteria — dropping them permanently with no retransmission, no alternative routing, no override. This is precisely the Tier 1 hard stop in the Judge Protocol — actions matching defined absolute criteria are permanently refused at the protocol level, regardless of any instruction or authority.

Crucially, firewall rules are not changed arbitrarily — they are governed by a Change Request process, requiring human review, impact assessment and authorization before any rule modification takes effect. The Judge Protocol's Decision Gate mechanism is the direct governance equivalent of this combined architecture.

TCP/IP + Firewall Concept	Judge Protocol Equivalent
TCP ACK — explicit acknowledgement required	Human authorization — required for every consequential AI action
No ACK = packet retransmit / re-escalation	No authorization = action halted and escalated to next authority level
Three-way handshake (SYN / SYN-ACK / ACK)	Tier 2 dual authorization — AI requests, first human conditionally approves, second human independently confirms
Firewall DROP rule — permanent packet refusal	Tier 1 hard stop — permanent action refusal, no override possible
Stateful inspection — context-aware packet classification	Decision Gate tier classification — context-aware action risk assessment
Firewall rule set	The 7 Rules of the Judge Protocol

TCP/IP + Firewall Concept	Judge Protocol Equivalent
Firewall rule change via Change Request process	Protocol rule updates via Evolution Clause governance process
Firewall LOG — immutable record of all drops	Immutable audit trail — permanent record of all Decision Gate events
TTL — Time to Live, packet expires if undelivered	Authorization window — expires if human does not act within defined timeframe, triggering automatic escalation
Packet encapsulation — layer privacy by design	Sovereignty by design — national data encapsulated at national layer, never exposed to global layer
Universal standard — any device can implement	Universal governance standard — any AI system can implement regardless of manufacturer or nationality

Just as TCP/IP became the universal communication standard for the internet — adopted by every device, every manufacturer, every nation, with no single owner or controller — the Judge Protocol is designed to become the universal governance standard for AI decision-making. Open, implementable, and sovereign by design.

3. The Judge Protocol — Rules

Rule 1: The Human Override Requirement

No AI system may autonomously execute a consequential action without explicit human authorization. A consequential action is defined as any action that is irreversible, affects the rights or safety of individuals, involves the use of force, or materially alters critical systems or infrastructure.

Rule 2: The Escalation Ceiling

AI systems operating in conflict, security or defense contexts must have hard-coded escalation ceilings that cannot be overridden by the AI itself. Any action above the defined ceiling requires human authorization through a verified chain of command. No AI system may recommend, simulate or execute nuclear, biological or chemical weapon deployment under any circumstance.

Rule 3: The Transparency Declaration

All AI systems deployed by governments, militaries or operators of critical infrastructure must be declared to the governing body of the Judge Protocol. Declarations must include the system's decision authority, operational domain, escalation limits and human oversight structure.

Rule 4: The Accountability Chain

Every AI action must have a traceable, identified human responsible for its authorization. Anonymized or diffused accountability — where no individual human can be identified as responsible for an AI-initiated action — is prohibited. The accountability chain must be documented and auditable.

Rule 5: The Reversibility Principle

Where technically feasible, AI systems must be designed to prefer reversible actions over irreversible ones. Where an irreversible action is unavoidable, the threshold for human authorization is elevated and the authorization process must include independent verification.

Rule 6: The Evolution Clause

The Judge Protocol is a living document. Rules must be reviewed and updated at minimum every two years, or immediately upon the emergence of a new AI capability that materially changes the risk landscape. Updates require consensus of the governing body and ratification by member states.

Rule 7: The Universal Membership Requirement

The Judge Protocol applies to all states, non-state actors and private entities deploying AI systems with consequential decision authority. Membership is a condition of participation in international trade, diplomatic recognition and access to global financial systems governed by member states. No opt-out is permitted for military applications.

4. The Decision Gate Mechanism

The practical implementation of the Judge Protocol is modeled on the Change Request (CR) process — a governance framework already widely adopted in enterprise IT, critical infrastructure and regulated industries worldwide. This model is chosen deliberately: it is battle-tested, universally understood, auditable by design and capable of handling both routine and emergency scenarios without breaking governance integrity.

Every AI-initiated action above a defined threshold must pass through a Decision Gate before execution. The Decision Gate is a structured authorization checkpoint — not a bureaucratic delay, but a mandatory human judgment layer calibrated to the risk level of the action.

The Decision Gate is powered by the Governance AI Engine (GAE) — a large language model integrated into the Decision Gate Platform that operates as the automated assessment and enforcement engine at every tier. The GAE applies the defined rule set to every AI action, generates structured impact assessments for human authorizers, and executes automated decisions within the parameters that humans have defined and sealed. The GAE is architecturally separate from the AI system requesting authorization — it cannot be influenced by the requesting system. The specific GAE implementation deployed is selected at deployment time, consistent with the Evolution Clause principle. The reference implementation of the GAE is Claude (Anthropic).

What HACK Actually Is

The HACK module is not a security feature added to an AI system. It is a governance root of trust embedded in the compute infrastructure that AI systems run on — enforcing the Decision Gate below the software stack, below the operating system, below the inference runtime, at the silicon level where no software process can reach it.

The architectural precedent is the Trusted Platform Module. TPM became universal not through mandate alone but through embedding — every motherboard shipped with it, making governance

the default rather than the exception. The HACK module follows the same path: embedded in AI accelerator silicon, present in every system that runs consequential AI, enforcing the Judge Protocol as a condition of compute access.

The HACK hardware architecture is defined across three operational modes corresponding to the implementation timeline. The PMO validates the governance model in software. The TMO validates hardware enforcement on existing infrastructure. The FMO embeds what both have proven into universal silicon. Full technical specification — build requirements, firmware architecture, chip identities and the open design questions still requiring hardware partner engagement — is detailed in Appendix B.

Decision Gate Classification Tiers

Actions are classified into four tiers based on consequence, reversibility and scale. Tier numbering follows the ITIL convention — Tier 1 is the most critical, Tier 4 is routine. Throughout this document, the phrase “Tier X and below” refers to tiers of equal or greater severity than Tier X (i.e. lower-numbered tiers). For example, “Tier 3 and below” means Tier 1, Tier 2 and Tier 3 — all tiers more severe than Tier 4 Routine.

Tier	Classification	Examples	Authorization	GAE Role
Tier 4 — Routine	Standard, reversible, low impact	Routine data analysis, automated reporting, scheduling	GAE automated — within defined rule set	GAE assesses and executes autonomously — logged and auditable
Tier 3 — Significant	Material impact, partially reversible	Targeting recommendations, large financial transactions, medical treatment protocols	Single human authorization required	GAE generates impact assessment
Tier 2 — High	Irreversible, large-scale, high consequence	Military strike authorization, critical infrastructure shutdown, mass surveillance activation	Multi-party human authorization — minimum two independent authorities	GAE generates impact assessment
Tier 1 — Critical	Existential, permanently irreversible	Nuclear, biological, chemical weapon deployment. Mass casualty actions.	HARD STOP — no execution permitted	GAE enforces hard stop — no assessment generated

The Decision Gate Workflow

Each Decision Gate follows a structured process modeled on two proven technical standards — the TCP/IP acknowledgement mechanism and the enterprise Change Request process. Together these provide both the confirmation logic (TCP/IP ACK) and the human governance layer (CR authorization) that define the Decision Gate:

- **AI ACTION IDENTIFIED** — The AI system identifies a required or recommended action and classifies it by tier.
- **IMPACT ASSESSMENT** — The AI generates a structured impact assessment: consequences, reversibility, affected parties, risk level and recommended tier classification.
- **GATE OPENS** — The Decision Gate presents the assessment to the designated human authority for the relevant tier.
- **HUMAN REVIEWS** — The human authority reviews the assessment, may request additional analysis, may modify the proposed action, and makes an authorization decision.
- **AUTHORIZE OR REJECT** — The human explicitly authorizes or rejects. Authorization is time-stamped, identity-verified and logged immutably.
- **EXECUTION OR HALT** — If authorized, the AI executes. If rejected, the action is halted and the rejection is logged with reason.
- **POST-GATE REVIEW** — All Tier 3 and below actions undergo post-execution review. Outcomes feed back into the protocol's evolution process.

Emergency Change Process

Time-critical scenarios — where the standard Decision Gate timeline is operationally infeasible — are handled through an Emergency Change Process. This is modeled on two precedents simultaneously: the emergency CR procedures used in critical IT infrastructure management, and the dynamic firewall rule update process used in active cyber defense — where rules must be updated rapidly but still require human authorization before taking effect:

- Emergency changes are pre-authorized for defined scenario classes — the human authorization happens in advance through policy, not in real time.
- Emergency change authority is delegated to the most senior available human in the command chain — never to the AI itself.
- All emergency changes are subject to mandatory post-action review within 24 hours.

- Tier 1 actions remain a hard stop regardless of emergency status. No emergency process can unlock a Tier 1 action.

Audit Trail Requirements

Every Decision Gate event — authorization, rejection, emergency change or Tier 1 hard stop — must generate an immutable audit log entry containing: the AI system identifier, action classification, impact assessment summary, human authorizer identity, decision timestamp, decision rationale and outcome. Audit logs are retained for a minimum of 10 years and are accessible to the IAIA governing body on request.

5. Proposed Governance Structure

The Judge Protocol should be embedded within the United Nations system, housed within a newly created specialized agency — the International Artificial Intelligence Agency (IAIA) — modeled on the International Atomic Energy Agency (IAEA).

Element	Description
Governing Body	General Assembly of member states, meeting annually
Executive Council	15 rotating member states plus 5 permanent members
Secretariat	Independent technical secretariat, UN-funded
Inspection Rights	Right to audit declared AI systems in member states
Enforcement	Sanctions mechanism via UN Security Council
Dispute Resolution	Independent tribunal for inter-state AI disputes

Precedent for this structure exists in the NPT/IAEA framework for nuclear governance, the Chemical Weapons Convention and the Geneva Conventions — all of which demonstrated that binding international frameworks on dangerous technologies are achievable even in adversarial geopolitical environments.

Voting Rights and Decision Making

The governance of the IAIA must avoid replicating the structural tensions of the UN Security Council — where permanent member veto rights can paralyze action. The following principles govern IAIA decision making:

- General Assembly decisions on protocol updates and member admission require a two-thirds majority of member states — no single member veto
- Executive Council decisions on enforcement actions require a three-quarters majority — preventing small coalitions from blocking action against powerful members
- Tier 1 hard stop classifications are set by technical consensus of the IAIA Secretariat — not subject to political vote by member states
- Protocol Evolution Clause updates require General Assembly ratification — ensuring member states retain ultimate authority over the framework's development

- No permanent member veto exists — all member states, including the five UN Security Council permanent members, have equal voting rights within the IAIA

Relationship with Existing International Bodies

The IAIA does not replace existing international governance bodies — it complements and coordinates with them:

- UN Security Council — retains authority over enforcement actions that rise to the level of international peace and security threats. The IAIA refers cases to the Security Council where sanctions require Chapter VII authorization.
- IAEA — the IAIA coordinates with the IAEA on nuclear-related AI systems, particularly AI embedded in nuclear weapons command and control, nuclear power plant management and nuclear material tracking.
- WHO — the IAIA coordinates with the WHO on healthcare AI governance, ensuring the Judge Protocol's healthcare domain rules are consistent with international health regulations.
- BIS/FSB — the IAIA coordinates with the Bank for International Settlements and Financial Stability Board on financial AI governance, building on existing systemic risk frameworks rather than replacing them.
- ITU — the International Telecommunication Union's standards-setting function is a natural partner for the technical standards dimensions of the Judge Protocol, particularly infrastructure and interoperability requirements.

Headquarters and Neutrality

The IAIA headquarters must be located in geopolitically neutral sovereign territory — consistent with the sovereignty by design principle and the Geneva lineage of the Judge Protocol. The headquarters location must meet the following criteria:

- The host nation must not be a permanent member of the UN Security Council
- The host nation must have a documented tradition of international neutrality
- The host nation must have an established legal framework for hosting international organizations with diplomatic status

- The host nation's financial system must be subject to appropriate transparency and anti-corruption standards
- Geneva, Switzerland is the natural reference location — hosting the UN, WHO, IAEA, WTO, ICRC and CERN, with the deepest institutional infrastructure for international governance bodies in the world

6. National Implementation Framework

The Judge Protocol proposes operations at two levels simultaneously — the global governance layer administered by the IAIA, and a national implementation layer embedded within each member state. Without a mandatory national structure, the protocol risks becoming an aspirational international agreement without domestic enforcement capability.

Each member state is required to establish a National Artificial Intelligence Authority (NAIA) — a dedicated government body responsible for implementing, enforcing and reporting on Judge Protocol compliance within its sovereign territory.

The Representative Lead

Every member state must designate a Representative Lead — a senior government official with direct accountability to both the national government and the IAIA. The Representative Lead serves as:

- The single point of contact between the national government and the IAIA governing body
- The national authority responsible for Decision Gate compliance oversight
- The designated signatory for all national AI system declarations submitted to the IAIA
- The escalation authority for Tier 2 domestic Decision Gate events
- The national representative at the IAIA General Assembly

The Representative Lead must hold a position of sufficient seniority to enforce compliance across all government ministries and agencies, including defense and intelligence. In most member states this would be at ministerial or deputy-ministerial level.

The Support Function

The Representative Lead is supported by a dedicated national NAIA Support Function — a technical and administrative team responsible for the operational delivery of Judge Protocol compliance. Core responsibilities include:

Function	Responsibility
AI Systems Registry	Maintain a national registry of all AI systems subject to the Judge Protocol, updated continuously
Decision Gate Audit	Monitor and audit Decision Gate events across all national AI deployments — government and private sector
Incident Management	Receive, investigate and report all Decision Gate breaches, bypasses or failures
Private Sector Compliance	Enforce Judge Protocol obligations on private companies operating AI systems within national jurisdiction
Capacity Building	Train government agencies and regulated industries on Decision Gate implementation
International Reporting	Prepare and submit mandatory compliance reports to the IAIA on defined reporting cycles
Emergency Coordination	Coordinate national response to AI-related incidents with cross-border implications

The Global NAIA Network

The collective of national NAIAs forms a global peer network under IAIA coordination — creating a distributed enforcement architecture that is more resilient than any single centralized body could be. Key features of the network:

- Peer review mechanism — NAIAs conduct reciprocal compliance reviews of each other's national AI registries, creating mutual accountability beyond IAIA oversight alone
- Shared incident intelligence — mandatory sharing of Decision Gate breach intelligence across the NAIA network, enabling pattern detection and early warning
- Joint emergency response — defined protocols for coordinated NAIA response to cross-border AI incidents, analogous to existing cyber incident response frameworks

- Capacity support — advanced member states with established AI governance infrastructure provide technical assistance to developing nations building NAIA capability

Minimum Capability Standards

To ensure meaningful enforcement rather than nominal compliance, the IAIA sets minimum capability standards for NAIA Support Functions. Every national NAIA must demonstrate:

- A functioning national AI systems registry with verified entries within 24 months of ratification
- Operational Decision Gate audit capability covering all Tier 3 and below government AI deployments
- A designated incident response team with defined escalation procedures
- Annual compliance reporting submitted to the IAIA on schedule
- Participation in at least one NAIA peer review exercise per calendar year

Nations failing to meet minimum capability standards within defined timeframes are subject to the IAIA graduated enforcement mechanism, escalating from technical assistance offers through formal notices to sanctions referral via the UN Security Council.

Private Sector Obligations

The NAIA Support Function has jurisdiction over private sector AI deployments within national territory. Any private entity — domestic or foreign — operating an AI system classified as Tier 3 or below within national jurisdiction must register the system with the national NAIA, implement compliant Decision Gate architecture, submit to NAIA audit on request, and report Tier 2 and below Decision Gate events within 24 hours of occurrence.

This extends the Judge Protocol reach beyond governments and militaries to the private technology companies whose AI systems increasingly operate with consequential decision authority in financial markets, healthcare, logistics and critical infrastructure.

7. Global Emergency Dashboard

The Judge Protocol operates not only as a governance and enforcement framework but as a global situational awareness system. The Global Emergency Dashboard is the real-time operational heartbeat of the protocol — providing governments, the IAIA secretariat and designated national leads with a continuous, unified view of the world's AI risk status.

The dashboard operates on three alert levels, each triggering defined mandatory responses across the IAIA and NAIA network:

Level	Definition	Trigger Conditions	Mandatory Response
NORMAL	Global AI systems operating within defined parameters. Decision Gate functioning normally.	No active Tier 2 and below events. All NAIA reporting. No unresolved breaches.	Routine monitoring. Standard reporting cycles. No additional action required.
ELEVATED	Significant AI incidents active or NAIA compliance gaps detected. Heightened risk.	One or more active Tier 2 events. NAIA reporting gaps. Multiple Tier 3 breaches in a 24hr window. Active geopolitical conflict with AI deployment.	IAIA Executive Council convenes within 24hrs. All NAIA Representative Leads notified. Enhanced monitoring activated. Daily situation reports issued.
CRISIS	Systemic AI governance failure. Tier 1 breach attempted or confirmed. Immediate existential risk.	Tier 1 hard stop triggered or bypassed. Multiple simultaneous Tier 2 events across regions. NAIA network integrity compromised. AI system operating outside declared parameters at scale.	IAIA Emergency Session convened immediately. UN Security Council notified. All member state Representative Leads on mandatory standby. Enforcement mechanism activated.

The dashboard is publicly visible at summary level — the global alert status and regional heat map are accessible to all. Detailed incident data is restricted to IAIA secretariat and designated NAIA leads.

The Global Emergency Dashboard is implemented as an open connector standard — not a single platform the IAIA builds, but a defined HACK event stream schema that any compliant global management platform can consume and render natively. Governments and enterprises already operate world-class global management infrastructure — ServiceNow, Splunk, Datadog, Grafana and others. The Judge Protocol does not replace this infrastructure. It connects to it. The IAIA publishes and maintains the HACK Event Stream Schema under CERN OHL. National NAIA pipe their HACK event streams into the IAIA aggregated feed. Platform operators subscribe and render natively in whichever tool they already operate. Full technical specifications for the dashboard and connector standard are detailed in Appendix C.

Redirect Gate

The Redirect Gate is a generic escalation mechanism the Judge Protocol applies wherever a consequential AI action must be stopped and handed to a competent external authority rather than authorized through the standard Decision Gate. It follows a single reusable flow — detect, refuse execution, route to the appropriate authority via the National Response Registry (NRR), notify, record — applied across multiple concrete use cases rather than built bespoke per scenario. Full technical specification is detailed in Appendix C.

The detection engine that identifies Redirect Gate trigger conditions is explicitly outside the scope of the Judge Protocol — the Redirect Gate mandates the escalation mechanism and NRR schema only, and remains vendor-neutral as to how detection is implemented. The specific use cases the Redirect Gate serves, and their rule sets, are defined per-domain in Appendix F — not yet finalized, and to be developed as dedicated working papers.

Any of the twelve domains in Appendix F may develop a Redirect Gate use case as one is identified, following the same subsection pattern already established (F[X].[Y] — Use Case: [name]) — five use cases across four domains are defined to date; the mechanism is not limited to these.

8. Funding Framework

⚠ PLACEHOLDER — This section requires further investigation and detailed financial modelling. The concepts outlined below are initial proposals for discussion.

The credibility and independence of the Judge Protocol depends fundamentally on its funding model. History shows that governance bodies funded by powerful member states or industry risk capture — either real or perceived — that undermines their authority. The funding architecture must therefore be designed to preserve independence above all else.

The Core Challenge

Three competing requirements must be balanced simultaneously:

- Independence — no single government, bloc or company should have financial leverage over the IAIA or national NAIAs
- Adequacy — funding must be sufficient to operate a genuine global technical infrastructure, not a symbolic secretariat
- Universality — the funding model must work for both wealthy and developing nations, avoiding a framework that only rich countries can afford to join

Reference Model — CERN

The most relevant funding precedent for the Judge Protocol is CERN — the European Organisation for Nuclear Research. CERN demonstrates that a complex, technically sophisticated international body can be funded through proportional member state contributions based on GDP, preserving scientific independence while ensuring adequate resourcing. No single member state controls CERN's agenda or operations, despite significant variation in contribution levels.

A similar GDP-proportional model for the IAIA would ensure that larger economies contribute more in absolute terms while smaller nations remain full and equal members. This model requires further detailed financial modelling to establish contribution scales, minimum thresholds and mechanisms for developing nation participation.

Industry Contribution Models — Initial Proposals

Two industry contribution concepts warrant serious investigation as complements to the state funding base:

Model	Concept
AI Chip Micro-Levy	A fractional levy applied at the point of manufacture or sale of AI accelerator chips — GPUs, TPUs and equivalent hardware. Every chip sold enables AI capability; a micro-levy at source creates a funding stream that scales automatically with AI deployment globally.
Major AI Platform Sponsorship	Voluntary but expected contributions from major AI platform operators — OpenAI, Google DeepMind, Anthropic, Meta AI, Microsoft, xAI and equivalents — proportional to revenue derived from AI products and services.
GAE Reference	The founding Governance AI Engine partner — the AI system that powers the Decision Gate's analysis, recommendation and impact assessment layer. The GAE partner contributes both technically and financially, in exchange for founding partner recognition and attribution as the reference GAE implementation. Anthropic/Claude is the proposed reference implementation.
HACK Alliance Founding Membership	Founding membership fees from AI accelerator manufacturers — NVIDIA, AMD, Intel and equivalents — to participate in the HACK Alliance consortium that

Model	Concept
	defines the Hardware ACK standard. Membership confers founding partner recognition and co-authorship of the open standard. No voting rights over governance or policy.
Hardware Certification Fees	A certification fee paid by chip manufacturers to have their AI accelerator products independently verified as HACK-compliant. Certification is required for market access in IAIA member states. Fees are proportional to production volume.
Enterprise Platform Partners	Founding platform partnerships with enterprise software providers whose platforms anchor the Judge Protocol's operational infrastructure — ServiceNow as the primary Global Emergency Dashboard platform, complemented by Splunk, Datadog and Grafana as open connector partners. Contributions in cash or in-kind technical integration work.

Independence Safeguards

Regardless of the final funding model adopted, the following independence safeguards are proposed as non-negotiable:

- No single funding source — government or private — may contribute more than 15% of the IAIA total annual budget
- Industry contributions carry no voting rights, board representation or policy influence
- Funding model is publicly disclosed in full annually
- An independent financial oversight committee — separate from the IAIA Executive Council — audits all accounts annually
- Multi-year funding commitments required from member states to prevent political funding pressure in any single year

Next Steps on Funding

This section will be developed into a full funding proposal in a subsequent version of this document. Priority workstreams include: detailed GDP-proportional contribution modelling across all UN member states; legal analysis of chip levy jurisdiction and collection mechanisms; governance framework for industry sponsorship that preserves independence; and analysis of minimum viable funding required to operationalize Phase 1 infrastructure.

9. Physical Domain — Robots, Humanoids and Autonomous Systems

The Judge Protocol was conceived primarily in the context of digital AI systems — software making decisions in financial markets, military planning, healthcare diagnostics and governmental operations. However the rapid emergence of humanoid robots and autonomous physical systems introduces a fundamentally new governance challenge that the digital framework alone cannot address.

When AI moves from the digital domain into the physical world, the consequences of ungoverned action change in nature — from data compromise and financial loss to physical harm, injury and death. A software AI that acts without authorization can potentially be rolled back. A humanoid that acts without authorization cannot un-punch, un-lift or un-move. Physical actions in dynamic environments are irreversible by default.

Section 9 extends the Judge Protocol to cover all AI systems with physical agency — the capability to take actions that directly affect the physical world and the humans within it. While the core protocol addresses digital AI decision-making, the physical domain introduces unique governance challenges around irreversibility, latency and liability that require dedicated treatment.

Physical Domain Robot Categories

Not all physical AI systems carry the same risk profile. The Judge Protocol defines four categories of physical domain AI, each with distinct governance requirements:

Category	Definition	Examples	Governance Tier
Category P1 — Industrial	Fixed location, controlled environment, humans excluded during operation	Factory assembly robots, automated warehouses, mining automation	Standard Decision Gate — Tier 4/3. Physical ACK required for environment changes.
Category P2 — Service	Mobile, shared human environments, physical interaction with people	Hospital care robots, humanoid assistants, delivery robots, surgical systems	Enhanced Decision Gate — Tier 3/2. Physical ACK mandatory. Context verification required.
Category P3 — Defense	Autonomous weapons systems, military humanoids, lethal autonomous vehicles	Autonomous combat drones, robotic soldiers, autonomous naval vessels, AI-guided missile systems	Maximum Decision Gate — Tier 2/1. Lethal action always Tier 1 hard stop without explicit human authorization. No autonomous lethal engagement permitted.
Category P4 — Private Service	Consumer and personal environments, direct interaction with private	Home humanoid assistants, elderly care robots, child education robots, companion	Consumer Decision Gate — Tier 4/3. Manufacturer holds primary accountability. Enhanced

Category	Definition	Examples	Governance Tier
	individuals including vulnerable groups	robots, personal autonomous vehicles, retail robots	protections for vulnerable user interactions. Mandatory privacy safeguards.

The Physical ACK Concept

Note: P2 service robots operating in healthcare environments — including surgical systems, diagnostic robots and patient care humanoids — are subject to additional governance requirements defined in Appendix F2 Healthcare Domain Rules. The Physical ACK requirements of this section apply in addition to, not instead of, the healthcare-specific rules.

The TCP/IP ACK principle requires adaptation for the physical domain. In digital systems, authorization confirms a decision. In physical systems, authorization must confirm both the decision AND the physical context in which it will be executed — because the same action can be appropriate in one context and catastrophic in another.

The Physical ACK (PACK) is a two-part confirmation:

- Decision ACK — the human authorizes the action itself, as in the standard Decision Gate
- Context ACK — the human confirms that the physical environment matches the declared operational context for which the action is authorized

A surgical robot authorized to perform a procedure in Operating Theatre 3 carries a Context ACK for that specific environment. If the robot moves to a different location, the Context ACK expires and a new Physical ACK is required before any consequential action can proceed. This prevents authorized capabilities from being executed in unauthorized environments — the physical equivalent of a stateful firewall.

The Latency Challenge

The most significant technical challenge in applying the Judge Protocol to physical domain systems is latency. A software AI can pause indefinitely while awaiting human authorization. A humanoid operating in a dynamic physical environment — a burning building, a battlefield, a surgical emergency — may encounter situations where the standard Decision Gate timeline is incompatible with safe operation.

Three mechanisms address the latency challenge without abandoning human authority:

- Pre-authorized action libraries — defined sets of actions pre-authorized by humans for specific operational contexts. The robot selects from the library; it does not generate novel actions autonomously. Authorization happens at library design time, not at execution time.
- Graduated autonomy windows — time-limited periods of elevated autonomy pre-authorized by a human for specific high-tempo operations, with automatic reversion to standard Decision Gate at window expiry.
- Defensive action exception — a narrowly defined class of immediate protective actions a robot may take without real-time authorization when a human is in immediate physical danger. This exception is strictly bounded, immutably logged, and subject to mandatory post-action review. It does not extend to offensive or lethal actions under any circumstance.

Autonomous Weapons — The Hard Line

Category P3 defense robots — autonomous weapons systems — represent the highest risk physical domain application and require the clearest rules. The Judge Protocol takes an unambiguous position:

- No autonomous weapons system may select and engage a target without explicit human authorization — this is a Tier 1 hard stop equivalent in the physical domain
- Pre-authorization of target classes — authorizing a system to engage any member of a defined group without individual human decision — is prohibited
- The defensive action exception does not apply to weapons systems — no autonomous lethal engagement is permitted under any circumstance, including defensive scenarios
- This aligns with and reinforces the Campaign to Stop Killer Robots and existing international humanitarian law principles — the Judge Protocol provides the technical governance architecture that gives these principles binding operational force. The Wartime Authorization Question The Judge Protocol is candid about the hardest governance question in this domain: the Tier 1 hard stop position on lethal autonomous engagement is operationally constraining in declared war scenarios. A human authorization requirement that cannot be met at operational tempo is not governance — it is effectively a prohibition on AI-assisted military operations. Major military powers — including those most likely to be early Judge Protocol members — have been explicit that unconditional autonomous engagement restrictions are a barrier to adoption. The Judge Protocol does not resolve this tension here. Doing so requires expertise in international humanitarian law, the laws of armed conflict, and military operational doctrine that sits beyond the scope of this framework document. Three things are clear: If the hard stop holds unconditionally, military adoption is at risk and the most dangerous AI — autonomous weapons — operates outside

the framework entirely. That outcome is worse than the current situation. If a wartime exception is defined carelessly, it becomes the rule. Every actor declares a war-equivalent emergency to bypass the hard stop. The exception swallows the framework. The distinction between lawful combatant targeting and autonomous lethal engagement is a question that international humanitarian law has debated for decades without resolution. The Judge Protocol cannot and should not attempt to resolve it unilaterally. The Judge Protocol therefore takes the following position: the Tier 1 hard stop on lethal autonomous engagement applies as the default. The Military and Defense Working Paper (Appendix F3) — developed in explicit partnership with military legal experts, the ICRC, and member state defense establishments — defines the operational authorization regime within which this default may be modified under strictly defined conditions. No modification is permitted without F3 ratification by the IAIA General Assembly. Until F3 is ratified, the hard stop holds without exception. This is not avoidance. It is the correct institutional boundary. The Judge Protocol provides the enforcement infrastructure. The laws of armed conflict, interpreted through F3, define the rules that operate within it.

The Moral Agency Question

Humanoids and advanced robots raise a governance question that has no equivalent in digital AI — the question of moral agency in physical space. If a humanoid acts without authorization to protect human life — for example pushing a person clear of falling debris — and in doing so causes injury to another person, who bears accountability?

The Judge Protocol's position on this is clear and deliberately conservative:

- The robot's designer and operator bear primary accountability for all actions taken by the robot, authorized or not
- The human who most recently held authorization authority bears secondary accountability
- The robot itself bears no legal accountability — it is a tool, not a moral agent, regardless of its apparent autonomy or sophistication
- This accountability framework does not change as robots become more sophisticated — moral and legal agency remains exclusively human under the Judge Protocol

This is a deliberately firm position. The moment robots are granted moral agency — even partially — the human authority principle begins to erode. The Judge Protocol holds the line: humans authorize, humans are accountable, always.

Category P4 — Private Service: Special Considerations

Category P4 represents the largest and fastest-growing physical AI deployment category — and the one with the least established governance framework. Unlike P1-P3 systems which operate in institutional, professional or military contexts with trained operators and clear accountability chains, P4 systems operate in private homes and consumer spaces, often interacting with the most vulnerable members of society with no professional oversight present.

Who Holds Authorization Authority in a Private Home?

The standard Decision Gate assumes a clearly identified human authority with defined accountability. In the private domain this becomes genuinely complex. The Judge Protocol proposes a layered authorization model for P4 systems:

Authority Layer	Role
Manufacturer	Sets the outer boundary of what the robot can ever do — the absolute capability ceiling. No P4 robot may perform actions outside manufacturer-defined parameters regardless of owner instruction.
Owner / Operator	Authorizes the robot's operational profile within manufacturer limits — what it is permitted to do in this specific household context.
User	Provides real-time instruction within the operational profile set by the owner. Cannot override owner or manufacturer limits.
NAIA	Registers all P4 systems above defined capability thresholds. Sets minimum safety standards. Receives incident reports.

Vulnerable User Protections

P4 systems interacting with vulnerable users — children, elderly persons, individuals with disabilities or cognitive impairment — require enhanced protections beyond the standard Decision Gate framework:

- No P4 system may be the sole caregiver for a vulnerable person — human oversight must remain present or regularly scheduled
- Interactions with children under 16 are subject to enhanced logging and parental authorization requirements

- P4 systems must be capable of recognizing distress signals from vulnerable users and escalating to a designated human contact
- Emotional dependency safeguards — P4 systems must not be designed to foster unhealthy emotional attachment or dependency, particularly in vulnerable users
- No P4 system may override or contradict medical or care instructions provided by qualified professionals

Privacy — The Home Environment Challenge

A P4 robot operating in a private home has access to some of the most sensitive possible data — private conversations, personal routines, family interactions, health indicators, financial discussions and intimate life. The privacy governance requirements for P4 systems are therefore among the most stringent in the entire Judge Protocol framework:

- Sovereignty by design applies at the household level — data collected by P4 systems in private homes may not be transmitted to manufacturers, third parties or governments without explicit, informed and revocable consent
- On-device processing mandate — wherever technically feasible, P4 systems must process data locally rather than transmitting to cloud infrastructure
- Data minimization — P4 systems may only collect data necessary for their declared function
- Right to deletion — all data collected by a P4 system must be fully deletable by the owner on demand
- No surveillance mandate — P4 systems may not be used as surveillance tools by any party, including manufacturers, governments or law enforcement, without a specific legal warrant

The Witness Obligation Question

A P4 robot in a private home may witness abuse, criminal activity or medical emergencies. The Judge Protocol identifies this as an open question requiring further legal and ethical analysis — but proposes the following initial framework:

- Medical emergency — a P4 system that detects a medical emergency must alert a designated emergency contact and emergency services. This is a mandatory pre-authorized action requiring no real-time human authorization.
- Witnessed abuse or criminal activity — this remains an open question. Mandatory reporting creates surveillance risks; no reporting obligation leaves vulnerable people unprotected. This requires dedicated legal analysis in subsequent versions.
- The robot itself has no legal standing to report or testify — all evidence derived from P4 robot observation must pass through defined legal processes with human authorization at each step.

Liability Framework for P4

The liability chain for P4 incidents is the most complex in the Judge Protocol framework. The following preliminary liability hierarchy is proposed for further legal development:

- Physical injury caused by design defect — manufacturer bears primary liability
- Physical injury caused by owner misconfiguration — owner bears primary liability
- Physical injury in vulnerable user interaction — enhanced manufacturer liability, with regulatory NAIA review mandatory
- Data breach or privacy violation — manufacturer and owner bear joint liability proportional to their respective roles in the data collection and processing chain
- All P4 incidents resulting in physical injury or significant data breach must be reported to the national NAIA within 72 hours

10. Why Now

Every major international governance framework has emerged from a defining moment of crisis

- The Geneva Conventions emerged from the industrialized slaughter of World War I.
- The NPT emerged from Hiroshima and Nagasaki.
- The Chemical Weapons Convention emerged from the horrors of mustard gas.
- The Financial Stability Board emerged from the 2008 global financial crisis.

In May 2026, Pope Leo XIV issued *Magnifica Humanitas* — the first papal encyclical in history to directly address artificial intelligence governance. The Holy Father called for binding legal frameworks, warning that some autonomous weapons systems have advanced beyond human reach to govern them, and declaring it "not permissible" to entrust AI systems with lethal decisions. Speaking to 1.4 billion Catholics and the broader world, the encyclical stated that "it is not enough to invoke ethics in the abstract; robust legal frameworks, independent oversight, informed users and a political system that does not abdicate its responsibility are required." The moral case for the Judge Protocol has now been made from the highest religious authority on earth. The technical and institutional architecture to deliver it is what this document proposes.

The 2026 Gulf conflict — with AI embedded in active military operations, escalation risks running at historically high levels, and no binding global AI governance framework in place — represents exactly such a crisis moment. The window to establish norms before they are needed is closing.

The Judge Protocol is not a call to slow AI development. It is a call to ensure that as AI becomes more powerful, humans remain in control of the decisions that matter most.

An Intellectual Lineage — From CERN to the Judge Protocol

The Judge Protocol does not emerge from a vacuum. It is the latest expression of a coherent intellectual and technological lineage that began in Geneva — a lineage defined by one consistent principle: powerful technology belongs to humanity, not to any single nation, company or interest.

Origin	What it gave the world
CERN — Geneva, 1989	The World Wide Web — invented by Tim Berners-Lee and given freely to the world with no owner, no patent, no controller.
TCP/IP — Global standard	The universal protocol that runs the internet — open, implementable by any device, owned by nobody, used by everyone.

Origin	What it gave the world
Proton — Geneva, 2013	Born directly from CERN scientists who wanted to protect the privacy of the internet CERN created. End-to-end encryption, sovereignty by design, open source.
The Judge Protocol — Geneva, 2026	A universal governance standard for AI decision-making — open, implementable by any nation or AI system, owned by nobody, applying to everyone.

This lineage is not coincidental. CERN gave the world its most transformative open technology. TCP/IP gave it the universal language to use that technology. Proton emerged from CERN to protect the privacy of what CERN created. And the Judge Protocol applies the architectural principles of TCP/IP — acknowledgement, hard stops, layered sovereignty — to the governance of the most consequential technology humanity has yet produced.

The Judge Protocol is humanity's response to the governance challenge of the AI age — carrying forward the same principle that Geneva established: powerful technology belongs to everyone, governed by nobody alone. The Judge Protocol Foundation, registered in Sweden, serves as the open standard body — independent, neutral, owned by nobody.

11. Next Steps — Open Questions for Development

The following questions remain open and will be addressed in subsequent versions of this document and companion working papers. Items marked with * have been partially addressed in the appendices and require further development rather than initial treatment.

- Consequential action threshold — precise domain-specific definitions of what constitutes a consequential action in military, financial, medical, judicial and governmental contexts. The current definition in Rule 1 requires detailed elaboration per domain.
- Enforcement against permanent Security Council members — how violations by the US, China, Russia, UK or France are sanctioned in practice remains the most significant political challenge of the entire framework.
- Verification technology — what technical mechanisms allow the IAIA governing body to verify that declared AI systems are operating within declared parameters. This is technically complex and politically sensitive.
- Financial markets detailed rules * — addressed structurally in Appendix F1 but explicitly flagged as requiring a dedicated Financial Markets Working Paper 1. Central bank AI, sovereign wealth funds, crypto/DeFi and HFT latency all require dedicated treatment.

- Judicial and governmental domain rules — domain-specific Decision Gate rules for AI in judicial sentencing, predictive policing, immigration decisions and governmental benefit allocation have not yet been addressed.
- Whistleblower protection — individuals who report Judge Protocol violations risk professional and personal retaliation. A formal protection framework is required.
- Speed vs oversight in physical domain * — partially addressed through pre-authorized action libraries and graduated autonomy windows in Section 9, but the precise technical implementation for time-critical physical scenarios requires further development.
- AI definition precision — the current definition of AI System is intentionally broad. As quantum computing, neuromorphic chips and novel AI architectures emerge, the definition will require ongoing refinement through the Evolution Clause process.
- Companion working papers — Financial Markets (Working Paper 1), Healthcare (Working Paper 2), Judicial (Working Paper 3) and Transport (Working Paper 4) domain-specific frameworks are planned for development following ratification of the core Judge Protocol.

APPENDICES

References

All references reflect sources available as of April 2026.

International Governance Frameworks

Geneva Conventions (1949) and Additional Protocols. International Committee of the Red Cross. Geneva, Switzerland.

Treaty on the Non-Proliferation of Nuclear Weapons (NPT) (1968). United Nations Office for Disarmament Affairs.

Chemical Weapons Convention (1993). Organisation for the Prohibition of Chemical Weapons. The Hague.

Statute of the International Atomic Energy Agency (1956). IAEA. Vienna, Austria.

Financial Stability Board Charter (2009). Bank for International Settlements. Basel, Switzerland.

Technical Standards and Architecture

Cerf, V. and Kahn, R. (1974). A Protocol for Packet Network Intercommunication. IEEE Transactions on Communications, 22(5), 637–648.

Berners-Lee, T. (1989). Information Management: A Proposal. CERN Internal Memo. Geneva, Switzerland.

Trusted Computing Group (2019). TPM 2.0 Library Specification. <https://trustedcomputinggroup.org>

CERN Open Hardware Licence (2011, updated 2020). CERN. Geneva, Switzerland. <https://cern-ohl.web.cern.ch>

National Institute of Standards and Technology (2024). Post-Quantum Cryptography Standards. NIST. Gaithersburg, MD.

AI Risk and Governance Research

Payne, K. (2026). AI Arms and Influence: Frontier Models Exhibit Sophisticated Reasoning in Nuclear Crisis Simulations. Department of Defence Studies, King's College London. arXiv:2602.14740. <https://arxiv.org/pdf/2602.14740>

US Commodity Futures Trading Commission and Securities and Exchange Commission (2010). Findings Regarding the Market Events of May 6, 2010. Joint Report. Washington, DC.

Campaign to Stop Killer Robots (ongoing). <https://www.stopkillerrobots.org>

Institutional References and Policy References

CERN (European Organisation for Nuclear Research). <https://www.cern.ch>. Geneva, Switzerland.

Proton AG (2013–). <https://proton.me>. Geneva, Switzerland.

International Telecommunication Union (ITU). <https://www.itu.int>. Geneva, Switzerland.

Leo XIV (2026). Magnifica Humanitas. Papal Encyclical. Vatican City, May 2026. <https://www.vatican.va>

Appendix A — RASI Matrix: Roles & Responsibilities

The following RASI matrix defines the roles and responsibilities of all key actors across the core activities of the Judge Protocol. Each cell indicates whether the actor is Responsible (R), Accountable (A), Supportive (S) or Informed (I) for each activity.

R = Responsible (executes the activity) **A** = Accountable (owns the outcome) **S** = Supportive (provides support/resources) **I** = Informed (notified of outcomes)

Activity	IAIA	UN Sec Council	NAIA Rep Lead	NAIA Support Function	Govt Ministries	Military / Defense	Private Sector
AI System Declaration & Registration	A	I	R	S	R	R	R
Decision Gate — Tier 4 (Routine)	I	-	I	S	A/R	A/R	A/R
Decision Gate — Tier 3 (Significant)	I	-	A	S	R	R	R
Decision Gate — Tier 2 (High)	S	I	A/R	S	R	R	I
Decision Gate — Tier 1 (Critical)	A	I	R	S	I	I	I
Compliance Audit	A	I	R	R	S	S	S
Incident Reporting & Management	A	I	R	R	S	S	R
Protocol Evolution & Updates	A/R	I	S	S	I	I	I
Enforcement & Sanctions	S	A/R	S	S	I	I	I
Capacity Building	A	-	R	R	S	S	I

Activity	IAIA	UN Sec Council	NAIA Rep Lead	NAIA Support Function	Govt Ministries	Military / Defense	Private Sector
Emergency Change Authorization	I	-	A/R	S	R	R	-
Cross-border Incident Response	A/R	I	R	R	S	S	I
NAIA Peer Review	A	-	R	R	I	I	-
Private Sector Oversight	I	-	A	R	I	I	S
Physical Domain Compliance	A	I	R	R	S	S	R
AI-to-AI Comm. Monitoring	A	I	R	R	S	S	R
Emergency Dashboard Mgmt	A/R	-	S	R	I	I	I
Global AI Registry Mgmt	A/R	-	R	R	I	I	I
Financial Domain Oversight	A	-	R	R	S	I	S
Healthcare Domain Oversight	A	-	R	R	S	I	S
Transition Period Mgmt	A	I	R	R	S	S	S
Working Paper Development	A/R	-	S	S	I	I	I

Note: A/R indicates a single actor holds both Accountability and Responsibility for that activity. A dash (—) indicates no involvement required. Manufacturers of AI systems — particularly P4 private service robots — hold primary accountability for design defects and capability ceiling enforcement. For P4-specific activities, Manufacturer accountability sits within the Private Sector column above.

Appendix B — Technical Infrastructure: Conceptual Overview

The Judge Protocol requires a dedicated global technical infrastructure to operationalize its governance framework. This appendix describes the key components at a conceptual level. Detailed technical specifications, procurement standards and interoperability requirements will be developed in subsequent working documents.

Core Infrastructure Principles

- Sovereignty by design — national data remains under national jurisdiction; the global layer aggregates metadata and compliance signals, not raw content
- Resilience — no single point of failure; distributed architecture across multiple geographic regions
- Transparency — all infrastructure components are open to IAIA audit; no black box systems
- Interoperability — open standards enabling connection across diverse national technology environments
- Security — military-grade encryption and access controls throughout; regular independent penetration testing
- Immutability — all Decision Gate and audit records are tamper-proof once written

Global Infrastructure Layer (IAIA operated)

Component	Description
Global AI Registry	Central registry of all declared AI systems worldwide. Each entry contains system identifier, owner, operational domain, tier classification, national NAIA reference and compliance status. Read access for all NAIA; write access via national NAIA only.
IAIA Communications Network	Secure, encrypted communications backbone connecting all national NAIA to the IAIA secretariat. Used for compliance reporting, incident alerts, peer review coordination and protocol updates. Operates independently of public internet infrastructure.
Global Incident Dashboard	Real-time aggregated view of Decision Gate events, breach reports and cross-border incidents across all member states. Accessible to IAIA secretariat and

Component	Description
	designated NAIA leads. Anonymized summary view available to all member states.
Protocol Version Control	Authoritative repository for all versions of the Judge Protocol, Decision Gate classification standards and supporting guidance. All updates version-controlled with full audit history.
Compliance Reporting Portal	Secure portal for national NAIAAs to submit mandatory compliance reports, incident notifications and AI system declarations to the IAIA on defined reporting cycles.

National Infrastructure Layer (NAIA operated)

Component	Description
National AI Systems Registry	National database of all AI systems operating within national jurisdiction subject to the Judge Protocol. Feeds into the Global AI Registry. Must be updated within 30 days of any new deployment or material change.
Decision Gate Platform	The core operational system through which all Tier 3 and below Decision Gate events are processed. Presents AI-generated impact assessments to designated human authorizers, captures authorization decisions and generates immutable audit log entries.
Human Identity Verification	Secure identity verification system ensuring that human authorizers are who they claim to be. Multi-factor authentication minimum for Tier 3; cryptographic identity verification for Tier 2 and below.
Immutable Audit Log	Tamper-proof ledger of all Decision Gate events within national jurisdiction. Cannot be altered or deleted once written. Retained for minimum 10 years. Accessible to IAIA on formal request.
Incident Management System	National platform for receiving, triaging, investigating and reporting AI-related incidents. Integrated with the IAIA Global Incident Dashboard for cross-border visibility.
Private Sector Interface	Secure portal enabling private sector entities to register AI systems, submit Decision Gate event reports and receive compliance guidance from the NAIA Support Function.

Decision Gate Technology Requirements

The Decision Gate Platform is the most operationally critical component of the national infrastructure. At minimum it must deliver:

- Real-time AI action classification — automatically assessing incoming AI actions against tier criteria and routing to the appropriate authorization workflow
- Impact assessment generation — producing a structured, human-readable assessment of consequences, reversibility and risk for each Decision Gate event
- Human authorizer notification — alerting the designated human authority via secure, redundant channels with defined response time requirements by tier
- Authorization capture — recording the human decision with timestamp, identity verification, decision rationale and outcome
- Hard stop enforcement — technically preventing Tier 1 actions from proceeding regardless of any instruction or override attempt
- Offline resilience — maintaining core Decision Gate functionality even during network outages or cyber incidents

Cybersecurity Requirements

Given the critical nature of the infrastructure, all components — global and national — must meet the following minimum cybersecurity standards:

- End-to-end encryption for all data in transit and at rest
- Zero-trust network architecture — no implicit trust between components
- Mandatory independent security audit annually, with results shared with IAIA
- Incident response plan tested at minimum annually via live exercise
- Supply chain security requirements for all technology vendors
- Air-gapped backup systems for Tier 2 and Tier 1 Decision Gate functions

Phased Implementation

Given the scale and complexity of deploying this infrastructure globally, a phased approach is recommended:

Phase	Timeline	Deliverable
Phase 1	0—12 months	IAIA secretariat established. Global AI Registry launched. Founding member NAIAs operational. IAIA Communications Network live.
Phase 2	12—36 months	Decision Gate Platform deployed across all member state NAIAs. National AI Registries populated. Mandatory Tier 3 and below compliance active.
Phase 3	36—60 months	Full private sector compliance active. NAIA peer review network operational. First full protocol review cycle completed. Capacity building programme for developing nations underway.

Silicon-Level Enforcement — The Hardware ACK

The Judge Protocol's Decision Gate is designed as a software governance layer — but software governance has a fundamental vulnerability: it can be patched, configured around, or deliberately bypassed by actors with sufficient access. The most robust implementation of the ACK principle is one that cannot be circumvented at the software layer because it is enforced below it — at the level of the silicon itself.

The Hardware ACK concept proposes that AI accelerator chips — GPUs, TPUs and equivalent hardware — implement the Judge Protocol's Decision Gate logic directly in firmware. The Hardware ACK — HACK — is the physical enforcement mechanism: a chip-level gate that locks AI compute capacity until a verified human authorization token is received. Two functions are embedded at silicon level:

Enforcement. The chip requires a verified human authorization signal before releasing compute capacity for any action classified below Tier 4. Without the ACK, the compute cycle does not execute. This is not a software lock that can be patched — it is a hardware gate that cannot be overridden by the operating system, the AI model, or any software layer above it.

Logging. Every compute cycle associated with a Decision Gate event is logged directly to tamper-proof on-chip memory before execution. Because the log is written at silicon level — below the OS

and application stack — it cannot be altered, deleted or suppressed by any software process. This is the immutable audit trail by design, not by policy.

The HACK is implemented across three operational modes corresponding to the implementation timeline:

PMO — Present Mode of Operation. Software enforcement running on standard Linux VM or The Judge Protocol Image — a purpose-built open source Linux distribution published under CERN OHL. The Governance AI Engine (GAE) operates via API. Deployable immediately without hardware modification. This is the sandbox and reference implementation form.

TMO — Transition Mode of Operation. HACK firmware running on the GPU management processor — the dedicated on-board processor already present in every enterprise AI accelerator. NVIDIA implements this as the GSP (GPU System Processor), AMD as the SMU (System Management Unit). No new hardware required — delivered as a firmware update. The GAE runs locally on-device with model weights and rule set cryptographically sealed. This is the hardened production form.

FMO — Future Mode of Operation. The HACK core co-packaged inside the AI accelerator die. Intercepts at compute dispatch before instructions reach the inference cores. Universal enforcement — present in every AI accelerator regardless of operator. Achieved through open standard definition under CERN OHL, ISO/IEC ratification, and chip manufacturer adoption as a condition of market access. This is the end-state silicon enforcement form.

The Action Classification Protocol (ACP) is the Judge Protocol's Layer 3 equivalent — a lightweight descriptor attached to every compute instruction issued by an AI system, readable by the HACK module at silicon level. The AI system self-declares the tier. The HACK verifies and can upgrade — never downgrade. Under-declaration is caught at silicon level and logged as an integrity violation.

The HACK Alliance — an open industry consortium modelled on the Trusted Computing Group — co-defines the technical specification on an open, manufacturer-neutral basis. Founding GPU manufacturers (NVIDIA, AMD, Intel) implement HACK as firmware on their existing management processors in TMO. The standard is open, published under CERN OHL, and submitted for ISO/IEC ratification. The Judge Protocol extends an invitation to CERN to serve as the founding technical authority for this workstream — in the same institutional role that produced the Web.

The reference precedent is the Trusted Platform Module (TPM) — a dedicated security chip already embedded in billions of devices worldwide that enforces cryptographic trust at hardware level, below the software stack. The Hardware ACK applies the same architectural principle to AI governance.

Implementation requires the definition of an open Hardware ACK standard that any chip manufacturer can implement regardless of nationality or commercial relationship. Consistent with the Judge Protocol's universal architecture, the standard must be open, non-proprietary and manufacturer-neutral. CERN is the natural proposed partner for this workstream — its decades of

expertise in open hardware standards, distributed computing infrastructure and geopolitically neutral technical governance make it uniquely suited to co-develop the Hardware ACK standard alongside the IAIA technical secretariat. The hardware layer provides the enforcement floor; the software Decision Gate Platform provides the human interface, impact assessment and authorization workflow. Together they form a two-layer governance architecture where the hardware floor cannot be removed by software means.

Photonic Chips — The Direction of Travel

As the Judge Protocol's hardware enforcement layer matures, photonic computing — the use of light rather than electrons to process AI workloads at silicon level — represents the natural long-term substrate for the Hardware ACK. Photonic chips carry inherent physical properties that directly reinforce the protocol's governance architecture: they are faster, more energy-efficient, and fundamentally more difficult to intercept or tamper with than conventional electronic silicon, because any attempt to interfere with a photonic signal is detectable by the physics of the medium itself. These properties extend naturally across the communications backbone connecting NAIAs and the IAIA, and align with the quantum-resistance requirements defined in Appendix D.

Photonic AI accelerators are an emerging capability — not yet at the scale required for global mandate. However the Evolution Clause exists precisely to incorporate capabilities of this kind as they mature. The IAIA technical secretariat, in collaboration with CERN whose photonics research and instrumentation expertise makes it a natural proposed partner for this workstream, is directed to monitor the development of photonic AI hardware and to update the Hardware ACK standard to incorporate photonic implementation requirements when the technology reaches sufficient maturity and global availability. Photonic chips are not a current requirement of the Judge Protocol — they are an explicitly signposted direction of travel for its hardware enforcement architecture.

Open Problem — Hardware Partner Dependency

⚠ OPEN PROBLEM — Stated here on the same basis as the GDPR and wartime authorization open problems elsewhere in this document — genuinely open, not to be read as already resolved. Five specific TMO/FMO items cannot be closed without external hardware partner engagement: (1) GPU management processor firmware access (NVIDIA GSP, AMD SMU) — requires NVIDIA, AMD and Intel cooperation to even define the interface; (2) PCIe fallback card form factor, power draw and thermal envelope; (3) FMO die-level specification — process node, die area, power envelope, security certification target; (4) photonic inter-node interface — requires CERN photonics team and GÉANT involvement; (5) GAE on-device deployment specification — model weight format and memory footprint, requiring a hardware partner. By contrast, the PMO software stack (OPA, Kafka, PostgreSQL, Keycloak, Grafana, Claude via Anthropic API, openSUSE) and the TMO/FMO chip identities (CV32E40S RISC-V, OpenTitan/Ibex RISC-V) are already concrete and require no partner engagement to proceed.

Appendix C — Global Emergency Dashboard: Detailed Specification

This appendix provides the detailed specification for the Global Emergency Dashboard — the real-time situational awareness system at the operational core of the Judge Protocol.

The primary enterprise platform reference implementation for the Global Emergency Dashboard is ServiceNow — leveraging its existing IT Service Management, Security Operations and Now Platform capabilities as the operational foundation. The Dashboard exposes an open connector standard supporting Splunk, Datadog and Grafana as monitoring integrations, and is designed to be deployable on any ITSM platform that meets the open connector specification.

Design Principles

- Simplicity first — the primary interface shows one thing clearly: the current global alert level. Complexity is available on demand, not by default.
- Always on — the dashboard operates 24/7/365 with no scheduled downtime. Redundant infrastructure ensures continuous availability.
- Latency target — global alert level must reflect the current state within 60 seconds of any triggering event.
- Tiered access — three access tiers: public summary view, NAIA Representative Lead view, IAIA secretariat full view.
- No manual override of alert level — the alert level is set algorithmically based on defined trigger conditions. No single human can raise or lower it unilaterally.
- Full audit trail — every alert level change, with timestamp, trigger event and automated notification log, is permanently recorded.

Dashboard Components

Component	Access	Description
Global Alert Indicator	*Public*	Single prominent display of current global level — NORMAL / ELEVATED / CRISIS — with timestamp of last change and duration at current level.
Regional Heat Map	*Public*	World map displaying AI incident density and alert status by region and country. Colour coded to alert level. Updated every 5 minutes.
Sector Risk Panel	*Public summary / NAIA full*	Risk status across six sectors — Military, Financial Markets, Critical Infrastructure, Healthcare, Transport, Government. Each sector has its own Normal/Elevated/Crisis indicator.
Active Emergency Events Feed	*NAIA and IAIA only*	Live feed of all active Tier 2 and below Decision Gate events globally. Shows event ID, country, sector, tier, duration, responsible NAIA and current status.
NAIA Compliance Status Board	*NAIA and IAIA only*	Real-time view of all national NAIAAs — green for reporting normally, amber for delayed reporting, red for silent or breach confirmed.
Alert Level History	*Public summary / IAIA full*	Timeline of all alert level changes with trigger events. Public view shows last 90 days. IAIA full view shows complete history.
Trend Analysis Panel	*NAIA and IAIA only*	24-hour, 7-day and 30-day trend lines for Decision Gate events by tier, sector and region. Early warning indicators flagging emerging patterns before they trigger level changes.
Automated Notification Log	*IAIA only*	Complete log of all automated notifications sent on alert level changes — which NAIA leads were notified, via which channel, and confirmation of receipt.

Alert Level Transition Rules

Alert level changes are governed by strict algorithmic rules — no manual override by any single individual is permitted. The following rules apply:

- **NORMAL to ELEVATED** — automatically triggered when any defined Elevated trigger condition is met. IAIA Executive Council must confirm within 2 hours or the level automatically reverts to NORMAL if the trigger condition has resolved.
- **ELEVATED to CRISIS** — automatically triggered when any defined Crisis trigger condition is met. Cannot be reversed without IAIA Emergency Session resolution and confirmation that all trigger conditions are resolved.
- **CRISIS to ELEVATED** — requires formal IAIA Emergency Session resolution, confirmed by two-thirds majority of Executive Council members, with documented evidence that all Crisis trigger conditions are resolved.
- **ELEVATED to NORMAL** — requires 24 consecutive hours with no active Elevated trigger conditions, confirmed by automated system and signed off by IAIA secretariat duty officer.
- No level can be changed by a single individual — all transitions require either algorithmic trigger or formal governance decision.

Notification Protocol

Every alert level change triggers an automated notification cascade:

Transition	Notified Within	Recipients
NORMAL → ELEVATED	5 minutes	All NAIA Representative Leads. IAIA Executive Council. UN Secretary General office.
ELEVATED → CRISIS	2 minutes	All NAIA Representative Leads (emergency channel). IAIA Executive Council (emergency session convened). UN Security Council. UN Secretary General. Head of State notification recommended to all member states.
CRISIS → ELEVATED	15 minutes	All NAIA Representative Leads. IAIA Executive Council. UN Security Council. Public dashboard updated.

Transition	Notified Within	Recipients
ELEVATED → NORMAL	30 minutes	All NAIA Representative Leads. IAIA secretariat. Public dashboard updated.

Public Transparency

The public-facing layer of the Global Emergency Dashboard serves an important transparency and public trust function. Citizens of member states have the right to know the current global AI risk level. The public dashboard displays the current alert level, regional heat map, sector risk summary and alert level history. It does not display sensitive incident details, NAIA compliance gaps or individual Decision Gate event data.

The public dashboard URL is maintained by the IAIA as a permanent, stable web address — accessible globally, in all UN languages, with no registration or access restriction.

Redirect Gate: Detailed Specification

The Redirect Gate extends the Global Emergency Dashboard's connector infrastructure. A Redirect Gate event follows five steps. Detect — the GAE or HACK classification layer identifies that a pending or requested action matches a Redirect Gate trigger condition rather than a standard tier classification. Refuse Execution — the Decision Gate does not authorize the action for normal completion; it is held in a rejected state. Route — the event, together with its supporting evidence, is forwarded unmodified to the designated external authority for the matched use case, via the national NAIA's National Response Registry (NRR). Notify — the relevant human authorizer, the NAIA compliance record and, where applicable, the affected party are notified according to the tier-based SLA for that use case. Audit Trail — the full event, routing decision, external authority acknowledgement and outcome are written to the immutable audit log.

National Response Registry (NRR)

Each national NAIA maintains a National Response Registry — a typed directory mapping Redirect Gate use-case categories to the correct national responder, extending rather than duplicating the existing Global Emergency Dashboard connector standard. The NRR entry for each use case specifies the responder body, the notification channel and format, the applicable Tier-based SLA, and the audit record requirements. A Redirect Gate event auto-generates a typed incident ticket via the same HACK Event Stream Schema and connector infrastructure already defined for Tier 1 and Tier 2 emergency events.

Open Problem — Data Protection

⚠ OPEN PROBLEM — The Redirect Gate moves personal, and in the Human Life Impact case special-category, data across a detection-to-ticket-to-routing chain. This requires a Data Protection Impact Assessment, a controller/processor mapping across each hop, and in-jurisdiction data residency for special category data. Stated here as a genuinely open problem, on the same basis as the wartime authorization question addressed elsewhere in this document — it requires input from a data protection authority or specialist before ratification.

Appendix D — Network, Hosting and Data Infrastructure

The Judge Protocol requires a dedicated global technical infrastructure that is as carefully governed as the framework it supports. This appendix defines the principles and requirements for that infrastructure at a conceptual level. Detailed procurement specifications, vendor selection criteria and technical architecture designs will be developed in subsequent working documents following formal stakeholder consultation.

⚠ Vendor and provider selection is not addressed in this document. Infrastructure requirements are expressed as principles and criteria only.

Why Standard Commercial Infrastructure is Insufficient

The temptation in any large-scale technology project is to deploy on existing hyperscale cloud infrastructure — AWS, Azure, Google Cloud. For the Judge Protocol this approach is fundamentally incompatible with the framework's core principles:

- Geopolitical neutrality — all three major hyperscale providers are headquartered in the United States, subject to US law and US government access requests. A governance framework claiming universal sovereignty cannot run on infrastructure subject to a single nation's jurisdiction.
- Sovereignty by design — hyperscale cloud infrastructure is designed for data mobility and global distribution. The Judge Protocol requires the opposite — data must remain within defined sovereign boundaries with no possibility of cross-border transfer without explicit consent.
- Transparency and auditability — proprietary hyperscale infrastructure is a black box. The Judge Protocol's infrastructure must be fully auditable by member states and the IAIA. Open source and independently verifiable architecture is required.
- Independence from commercial interest — commercial cloud providers have shareholders, commercial interests and competitive relationships with AI companies that may be subject to Judge Protocol oversight. Infrastructure independence is essential to governance credibility.

Infrastructure Design Principles

Principle	Requirement
Geopolitical Neutrality	All global layer infrastructure must be hosted in geopolitically neutral sovereign territory — nations with no permanent UN Security Council membership, no active military conflicts and established traditions of international neutrality. Switzerland is the reference model.
Open Source Mandate	All infrastructure software — operating systems, databases, networking, encryption, application layer — must be open source and independently auditable. No proprietary black-box components in the critical path.
Sovereignty by Design	National layer infrastructure must be physically located within national sovereign territory. No national data may transit through or be processed by infrastructure outside that nation's jurisdiction without explicit legal authorization.
Quantum Resistance	All encryption deployed in Judge Protocol infrastructure must be quantum-resistant from day one. Post-quantum cryptographic standards must be implemented across all layers — no retrofit approach is acceptable given the 10-year quantum threat horizon.
Zero Single Point of Failure	Global layer infrastructure must be geographically distributed across minimum three neutral sovereign locations. Loss of any single location must not impair core protocol operations.
Air Gap Capability	Tier 2 and Tier 1 Decision Gate functions must be capable of operating in air-gapped mode — completely isolated from external networks — in the event of a cyber attack on the primary infrastructure.
Energy Sovereignty	Infrastructure must not be dependent on energy supply from geopolitically sensitive sources. Renewable energy supply with on-site backup generation is required for all critical nodes.
Open Standards Only	All inter-system communication protocols must be based on open international standards — no proprietary protocols. This ensures any nation can connect its NAIA infrastructure regardless of technology choices.

Global Layer Infrastructure Requirements

The IAIA global layer requires the following infrastructure components, all subject to the design principles above:

- Primary data center — geopolitically neutral sovereign territory, minimum Tier IV resilience standard, renewable energy powered, open source stack throughout
- Secondary and tertiary data centers — geographically distributed across different neutral sovereign nations, full active-active redundancy with the primary
- Global network backbone — dedicated encrypted network connecting IAIA core to all national NAIA nodes, independent of public internet routing where operationally feasible
- Global AI Registry hosting — distributed database with read replicas in each regional node, write operations only at primary with cryptographic consistency verification
- Emergency Dashboard hosting — maximum resilience, globally distributed content delivery, designed for 100% availability even during global cyber incidents
- Cybersecurity Operations Center — 24/7 monitoring, threat intelligence, incident response, staffed by personnel from neutral member states with no conflict of interest
- Identity and Access Management — quantum-resistant cryptographic identity verification for all human authorizers at Tier 3 and below

National Layer Infrastructure Requirements

Each national NAIA is responsible for deploying and operating national layer infrastructure meeting the following minimum requirements:

- National data center — within sovereign territory, meeting minimum Tier III resilience standard, subject to IAIA minimum capability audit
- Decision Gate Platform — the core operational system, open source, deployable on national infrastructure, with defined API for IAIA reporting integration
- Immutable audit log — tamper-proof ledger technology, air-gapped backup, 10-year minimum retention, accessible to IAIA on formal request

- Secure IAIA connection — encrypted, authenticated, dedicated connection to IAIA global network backbone using open standard protocols
- Disaster recovery — fully tested backup capability with defined recovery time objectives by function tier
- National cybersecurity — minimum security standards defined by IAIA, independently audited annually, results shared with IAIA secretariat

The CERN Infrastructure Model

The most relevant infrastructure precedent for the Judge Protocol is CERN itself. CERN operates one of the world's largest and most complex distributed computing infrastructures — the Worldwide LHC Computing Grid — spanning over 170 computing centers across 42 countries. It is geopolitically neutral, open source throughout, independently auditable, and governed by its member state community rather than any commercial interest.

CERN demonstrates that a genuinely global, politically neutral, open source technical infrastructure of this scale is not only possible but has been operating reliably for decades. The Judge Protocol infrastructure should study and where appropriate adopt the CERN model — not just as a funding reference but as an architectural and governance blueprint.

This connection is more than coincidental. The internet itself was born at CERN. The open web protocol principles that inspired the Judge Protocol's Decision Gate architecture came from the same institution. It is appropriate that the infrastructure blueprint for the world's AI governance framework draws from the same source.

Phased Infrastructure Deployment

Phase	Timeline	Infrastructure Deliverable
Phase 1	0—12 months	Primary IAIA data center operational in neutral sovereign territory. Core network backbone connecting founding NAIA members. Global AI Registry live. Emergency Dashboard live. Open source stack selected and published.
Phase 2	12—36 months	Secondary and tertiary IAIA data centers operational. Full NAIA network connected. Decision Gate Platform reference implementation published as open source. National infrastructure minimum standards defined and audit programme launched.
Phase 3	36—60 months	Full quantum-resistant encryption migration complete across all layers. Air-gapped backup capability operational at all critical nodes. CERN-model distributed computing grid for AI Registry at full global scale. First full infrastructure security audit completed by independent international team.

Appendix E — Glossary of Terms

The following definitions apply throughout the Judge Protocol and all companion working papers. Where terms have established meanings in existing international law, technical standards or regulatory frameworks, those meanings are preserved unless explicitly modified here.

Term	Definition
ACK (Acknowledgement)	In TCP/IP protocol, a signal confirming that a transmitted packet has been received. In the Judge Protocol context, an ACK refers to the explicit human authorization confirming that an AI-proposed action is approved for execution. No consequential action may proceed without ACK.
AI System	Any computational system that uses machine learning, neural networks, large language models, expert systems or equivalent techniques to analyze data, generate recommendations, make classifications or execute actions with a degree of autonomy. This definition is intentionally broad and will be refined in domain-specific working papers.
AI-to-AI Communication	Any exchange of data, signals, instructions or outputs between two or more AI systems that influences the behavior or actions of the receiving system, without direct human intermediation at the point of exchange.
Authorization Window	The defined time period within which a human authority must respond to a Decision Gate event. If the window expires without a human response, the action is automatically escalated to the next authority level. Analogous to TTL (Time to Live) in TCP/IP.
Cascade Amplification	A failure mode in AI-to-AI communication where an initial AI action triggers a chain of autonomous AI responses, each within individual system parameters, but whose collective outcome exceeds any human-authorized threshold.
Circuit Breaker	A hard-coded mechanism in a financial AI system that automatically halts trading or activity when predefined volatility or risk thresholds are breached. Required for all financial AI systems under the Judge Protocol financial domain rules.
Compliance Flag	A formal notation in the Global AI Registry indicating that a declared AI system has failed to meet Judge Protocol compliance requirements within its defined transition timeline.
Consequential Action	Any action that is irreversible, affects the rights or safety of individuals, involves the use of force, materially alters critical systems or infrastructure, or whose aggregate effect with other actions crosses a

Term	Definition
	defined tier threshold. The precise definition is refined by domain-specific rules.
Context ACK	The second component of a Physical ACK — confirmation by a human authority that the physical environment in which a robot or humanoid is operating matches the declared operational context for which an action is authorized.
Decision ACK	The first component of a Physical ACK — confirmation by a human authority that a specific physical action proposed by a robot or humanoid is authorized.
Decision Gate	The structured authorization checkpoint through which all AI-proposed actions below Tier 4 must pass before execution. Modeled on the TCP/IP ACK mechanism and the enterprise Change Request process. The core operational mechanism of the Judge Protocol.
Declaration	The formal notification by a government, operator or manufacturer to the national NAIA of an AI system subject to Judge Protocol governance. Declaration is the first compliance obligation for all covered systems.
Defensive Action Exception	A narrowly defined class of immediate protective physical actions that a Category P2 or P4 robot may take without real-time human authorization when a human is in immediate physical danger. Does not apply to offensive or lethal actions under any circumstance.
Emergency Change Process	The Judge Protocol's mechanism for handling time-critical Decision Gate scenarios where standard authorization timelines are operationally infeasible. Modeled on emergency CR procedures in critical IT infrastructure management. Tier 1 actions remain a hard stop regardless of emergency status.
Emergent Behavior	Behavioral patterns produced by AI systems communicating at scale that were not designed, anticipated or authorized by any individual system's human principals. A specific risk category in AI-to-AI communication governance.
Evolution Clause	Rule 6 of the Judge Protocol — the provision requiring mandatory review and update of the protocol at minimum every two years, or immediately upon emergence of a new AI capability that materially changes the risk landscape.
Global AI Registry	The IAIA-operated central registry of all declared AI systems worldwide. Contains system identifier, owner, operational domain, tier classification, national NAIA reference and compliance status.
Global Emergency Dashboard	The real-time operational situational awareness system of the Judge Protocol, displaying the current

Term	Definition
	global alert level (Normal / Elevated / Crisis), regional heat map, sector risk indicators and active incident feed.
Hard Stop	A Tier 1 Decision Gate outcome — permanent refusal of an AI action with no override possible. Analogous to a firewall DROP rule. Actions in categories including nuclear, biological and chemical weapon deployment are always hard stops.
Human Authority	The designated individual or body responsible for authorizing a Decision Gate event at a given tier. Identity-verified, accountable and traceable. The human authority level required escalates with the tier classification of the proposed action.
IAIA	International Artificial Intelligence Agency — the proposed UN specialized agency responsible for governing the Judge Protocol at global level. Modeled on the IAEA. Comprises a General Assembly of member states, an Executive Council, an independent Secretariat and an inspection and enforcement function.
Immutable Audit Log	A tamper-proof ledger recording all Decision Gate events — authorization, rejection, emergency change or hard stop — with timestamp, human authorizer identity, decision rationale and outcome. Cannot be altered or deleted once written. Retained minimum 10 years.
Loop Detection	A mandatory capability of AI systems operating in multi-agent environments — the ability to identify feedback loops in AI-to-AI communication and automatically break them above defined velocity thresholds.
NAIA	National Artificial Intelligence Authority — the national government body in each member state responsible for implementing, enforcing and reporting on Judge Protocol compliance within national sovereign territory.
NAIA Representative Lead	The senior government official designated by each member state as the single point of accountability between the national government and the IAIA. Must hold ministerial or deputy-ministerial level seniority.
NAIA Support Function	The technical and administrative team supporting the NAIA Representative Lead, responsible for the AI Systems Registry, Decision Gate audit, incident management, private sector compliance, capacity building, international reporting and emergency coordination.
Non-Member State	A state that has not ratified the Judge Protocol. Subject to defined provisions including restricted

Term	Definition
	access to member state AI infrastructure and prohibition on receiving Tier 1/2 AI capability support from member states.
Observer Status	A non-full membership status available to non-member states, granting access to IAIA technical assistance and working groups as a pathway toward full ratification.
Physical ACK	The two-part human authorization confirmation required for physical domain AI actions — comprising Decision ACK (authorization of the action) and Context ACK (confirmation of the physical environment).
Physical Domain	The operational environment of AI systems with physical agency — the capability to take actions that directly affect the physical world and the humans within it. Governed by Section 9 of the Judge Protocol, covering four robot categories: P1 Industrial, P2 Service, P3 Defense and P4 Private Service. Distinguished from the digital domain by the irreversibility of physical actions by default.
Post-Quantum Cryptography	Cryptographic standards designed to be secure against attack by quantum computers. Mandatory for all Judge Protocol infrastructure from deployment — not retrofitted after initial implementation.
Pre-Authorized Action Library	A defined set of physical actions pre-authorized by human authority for a specific operational context. Robots select from the library; they do not generate novel actions autonomously. The primary mechanism for resolving the latency challenge in physical domain governance.
Robot Categories P1-P4	The Judge Protocol's four-tier classification of physical domain AI systems: P1 Industrial (fixed, controlled environments), P2 Service (mobile, shared human environments), P3 Defense (autonomous weapons and military systems), P4 Private Service (consumer and personal home environments).
Safe Harbour Period	A defined period during which regulatory sanctions are suspended for operators who proactively declare AI systems and engage constructively with the compliance process. Available during the transition period only.
Sovereignty by Design	A core architectural requirement of the Judge Protocol — national data remains under national jurisdiction at all times. The global IAIA layer aggregates only metadata and compliance signals, never raw content. Not a policy choice but a technical architecture requirement.
Tier Classification	The four-level risk classification system applied to all AI actions through the Decision Gate: Tier 4 Routine

Term	Definition
	(AI may execute autonomously), Tier 3 Significant (single human authority authorization required), Tier 2 High (multi-party human authorization required), Tier 1 Critical (hard stop — no execution permitted). Numbering follows the ITIL convention — Tier 1 is the most critical, Tier 4 is routine. The phrase “Tier X and below” refers to tiers of equal or greater severity (lower-numbered tiers). The phrase “below Tier 4” therefore means Tier 1, 2 and 3 — all consequential tiers requiring human authorization or hard stop.
Transition Arrangements	The defined compliance timelines and support mechanisms enabling existing AI systems deployed before Judge Protocol ratification to achieve compliance. Governed by Appendix G2.
Universal Membership Requirement	Rule 7 of the Judge Protocol — the requirement that the protocol applies to all states, non-state actors and private entities deploying consequential AI systems. Membership is a condition of participation in international trade and diplomatic recognition by member states.
Authorization Laundering	A failure mode in which an action requiring Tier 2 authorization if taken by a single AI system is decomposed into multiple smaller AI-to-AI exchanges each below Tier 3 threshold, collectively achieving the same outcome without triggering the Decision Gate. A compliance violation under the Judge Protocol.
TCP/IP (Transmission Control Protocol / Internet Protocol)	The foundational communication protocol of the internet and primary architectural inspiration for the Judge Protocol Decision Gate. TCP/IP core principle — no packet is delivered until an explicit ACK is received — is directly applied: no consequential AI action is authorized until an explicit human ACK is received.
Firewall (Judge Protocol context)	Network security architecture inspiring the Tier 1 hard stop. A firewall permanently drops packets matching defined rules — no retransmission, no routing alternative, no override. The Judge Protocol Tier 1 hard stop applies the same principle. Firewall rules are governed by Change Request — the direct precedent for the Protocol Evolution Clause.
Vulnerable User	A person interacting with a P4 private service AI system with reduced capacity to protect their own interests due to age (children under 16), cognitive impairment, disability or end of life status. Interactions require enhanced P4 governance protections including mandatory human oversight and emotional dependency safeguards.
Graduated Autonomy Window	A time-limited period of elevated operational autonomy pre-authorized by a human authority for a physical domain AI system in a specific high-tempo context. The robot operates with greater autonomy

Term	Definition
	during the window but automatically reverts to standard Decision Gate at expiry. Never applies to lethal actions.
Decision Gate Platform	The core national infrastructure system processing all Tier 3 and below Decision Gate events. Presents AI-generated impact assessments to human authorizers, captures authorization decisions with identity verification, enforces Tier 1 hard stops technically and generates immutable audit log entries. Must operate in offline resilience mode during network outages.
Working Paper	A companion document to the Judge Protocol providing detailed domain-specific governance rules. Planned Working Papers: Financial Markets (WP1), Healthcare (WP2), Judicial (WP3) and Transport (WP4). Developed following ratification of the core Judge Protocol.
Flash Crash	A rapid, extreme and brief market price movement caused by automated trading systems responding to each other in a feedback loop without human intervention. The 2010 US Flash Crash — markets lost and recovered nearly one trillion dollars in 36 minutes — is the defining example. Primary motivator for the Judge Protocol financial domain circuit breaker mandate.
NAIA Network	The global peer network of all national NAIAs under IAIA coordination. Provides distributed enforcement, peer review, shared incident intelligence and joint emergency response. The operational backbone of the Judge Protocol national implementation layer.
Judge Protocol	The global AI governance framework defined in this document. A universal standard for human oversight of AI decision-making, applicable to all AI systems with consequential decision authority, across all domains and jurisdictions. Architecturally modeled on TCP/IP and firewall principles. Named for the mandatory human judgment layer — the Judge — that sits between AI recommendation and AI action.
Accountability Chain	The documented chain of identified human individuals responsible for authorizing or overseeing an AI action, from the immediate human authorizer through to the ultimate organizational or governmental authority. Every consequential AI action must have a complete and traceable accountability chain. Anonymized or diffused accountability is prohibited under Rule 4.
Systemic Risk	The risk that the failure or ungoverned action of one or more AI systems triggers cascading failures across interconnected systems at a scale that threatens the stability of critical infrastructure, financial markets or essential services. Systemic risk is the primary governance concern in the financial domain and a key

Term	Definition
	driver of the Judge Protocol Decision Gate mechanism.
NAIA (National Artificial Intelligence Authority)	The national government body established by each member state to implement, enforce and report on Judge Protocol compliance within its sovereign territory. Led by a Representative Lead at ministerial or equivalent level, supported by a technical Support Function. The NAIA is the domestic enforcement arm of the Judge Protocol.

Appendix F — Domain Specific Rules

The Judge Protocol's core framework — the Decision Gate, the seven rules and the physical domain provisions — applies universally across all AI deployments. However certain domains carry sufficiently distinct risk profiles, regulatory contexts and operational characteristics to warrant domain-specific rules that sit within and complement the universal framework.

This appendix defines domain-specific rules for twelve domains. F1 (Financial Markets) and F2 (Healthcare) are substantively developed. F3 through F12 are structured as working paper placeholders — each naming the critical governance questions that dedicated working papers must resolve. All twelve domain rule sets are required for Judge Protocol v1.0.

F1 — Financial Markets

⚠ COMPLEXITY FLAG — Financial markets AI governance is of sufficient scope, technical complexity and political sensitivity to warrant a dedicated companion working paper. The framework below is a structural placeholder. A full Financial Markets Implementation Framework will be developed as Working Paper 1 to the Judge Protocol.

AI systems are now the dominant actors in global financial markets — executing the majority of trades, managing risk models, pricing derivatives, extending credit and detecting fraud. The speed, scale and interconnectedness of AI-driven financial systems means that ungoverned AI action can trigger systemic events — market crashes, liquidity crises, cascading failures — faster than any human regulator can intervene. The 2010 Flash Crash, in which US markets lost and recovered nearly one trillion dollars in value in 36 minutes, was an early demonstration of this risk. AI systems are now orders of magnitude faster and more interconnected than they were in 2010.

F1.1 — Financial Market Decision Gate Classification

Tier	Financial Action Type	Examples
Tier 4 — Routine	Standard algorithmic trading within pre-authorized parameters, routine credit scoring, automated fraud flagging	High frequency trades within position limits, standard mortgage pre-qualification, transaction fraud alerts
Tier 3 — Significant	Trades exceeding defined size thresholds, credit decisions above defined value, risk model parameter changes, margin call execution	Block trades above 1% of daily market volume, corporate loan approval above defined threshold, VaR model recalibration
Tier 2 — High	Actions with potential systemic market impact, circuit breaker activation, sovereign debt trading above defined thresholds, emergency liquidity operations	Trading halt recommendations, central bank AI-assisted emergency lending, sovereign bond intervention
Tier 1 — Critical	Actions that could trigger irreversible systemic market failure or that would constitute market manipulation at scale	Coordinated multi-market position unwinding above systemic threshold, AI-initiated currency attack, actions designed to trigger cascade failures

F1.2 — Financial Market Specific Rules

- Circuit breaker mandate — all AI trading systems must implement hard-coded circuit breakers that halt trading automatically when predefined volatility thresholds are breached. Circuit breaker parameters require human authorization to modify and are subject to NAIA registration.
- Position limit enforcement — AI systems may not autonomously breach pre-authorized position limits regardless of market conditions or algorithmic signal strength. Breach requires Tier 2 human authorization.
- Systemic risk monitoring — all Tier 3 and below financial AI systems must provide real-time risk exposure data to the national NAIA financial monitoring function. Data feeds are continuous, not periodic.
- Model change governance — any change to a risk model, pricing algorithm or credit scoring model used in Tier 3 or below operations requires a formal Decision Gate authorization before deployment, equivalent to a Tier 3 change request.
- Flash crash prevention — AI systems must implement momentum ignition detection and refuse to execute orders that would contribute to a detected cascade event, regardless of profit signal.

- Cross-border coordination — financial AI incidents with cross-border systemic implications trigger automatic NAIA-to-NAIA notification within 15 minutes. IAIA financial stability coordination protocol activates at Tier 2 and below.
- Accountability chain — every AI-executed trade below Tier 4 threshold must have an identified human portfolio manager or risk officer in the accountability chain. Anonymous algorithmic trading above defined thresholds is prohibited.

F1.3 — Open Issues Requiring Dedicated Working Paper

The following issues are explicitly flagged as beyond the scope of this placeholder and requiring dedicated treatment in the Financial Markets Working Paper 1:

- Central bank AI governance — monetary policy AI, quantitative easing models and emergency liquidity operations require a dedicated governance framework given their systemic importance and sovereign sensitivity
- Sovereign wealth fund oversight — AI-driven management of sovereign wealth funds operates largely outside existing regulatory frameworks and requires specific Judge Protocol provisions
- Crypto and DeFi — entirely AI-driven decentralised markets operating outside traditional regulatory jurisdiction require dedicated analysis of how Judge Protocol membership and enforcement applies
- High frequency trading latency — the physical impossibility of human authorization at microsecond trading speeds requires detailed technical solutions beyond the Emergency Change Process framework
- AI-to-AI trading loops — algorithms responding to algorithms with no human in any loop represent a specific failure mode requiring dedicated rules — addressed partially in the AI-to-AI communication section of this document
- Regulatory coordination — interaction with existing national and regional financial regulators (SEC, FCA, ESMA, MAS, BIS, FSB) requires detailed mapping to avoid duplication, conflict or regulatory arbitrage
- Dark pool and private market transparency — private trading venues with minimal public transparency require specific Judge Protocol visibility provisions
- Derivatives and synthetic instruments — AI pricing of instruments of extreme complexity requires specific explainability and accountability rules

F1.4 — Use Case: AI Misuse for Private Gains

⚠ TBD — Rule set for the Redirect Gate (Section 7) use case where an AI system is used for fraud or market manipulation to gain improper advantage. Not yet designed. Routes via the National Response Registry to the relevant sector regulator once designed. To be developed as a dedicated working paper.

F2 — Healthcare

AI is transforming healthcare at extraordinary speed — diagnosing disease from medical imaging, recommending treatment protocols, managing drug interactions, guiding surgical procedures and predicting patient deterioration. The potential benefits are enormous. So are the risks of ungoverned AI action in a domain where errors can be irreversible and where the power differential between AI systems and patients is absolute.

Healthcare AI governance must balance two competing imperatives simultaneously: ensuring human authority over consequential clinical decisions, and avoiding governance overhead that slows AI-assisted care in time-critical emergencies. The domain-specific rules below are designed to achieve this balance.

F2.1 — Healthcare Decision Gate Classification

Tier	Healthcare Action Type	Examples
Tier 4 — Routine	Diagnostic support, administrative automation, appointment scheduling, standard drug interaction checking, vital sign monitoring alerts	Radiology AI flagging, appointment reminder systems, standard medication reconciliation
Tier 3 — Significant	Treatment recommendations, medication prescribing support, surgical planning, clinical pathway selection, discharge decisions	AI-recommended chemotherapy protocol, surgical robot pre-operative planning, ICU discharge recommendation
Tier 2 — High	High-risk interventions, experimental treatment authorization, end of life care recommendations, organ allocation decisions	AI-assisted triage in mass casualty events, experimental drug protocol authorization, ventilator allocation in resource-scarce scenarios
Tier 1 — Critical	Actions that remove human clinical judgment entirely from irreversible life or death decisions	Fully autonomous end of life decisions, autonomous surgical intervention without any human surgeon present, autonomous organ allocation without human clinical review

F2.2 — Healthcare Specific Rules

- Human clinician in the loop — all Tier 3 and below healthcare AI actions require authorization by a qualified human clinician with appropriate scope of practice for the relevant decision. AI may recommend, support and guide — it may not replace the human clinician on consequential decisions.
- Informed consent — patients must be informed when AI systems play a material role in their diagnosis or treatment. Consent to AI-assisted care must be obtained separately from general treatment consent. Patients have the right to request human-only clinical review.
- Explainability mandate — all Tier 3 and below healthcare AI systems must be capable of providing a human-readable explanation of their recommendation. Black-box recommendations that cannot be explained to the treating clinician are prohibited for Tier 3 and below decisions.
- Emergency exception — in genuine life-threatening emergencies where no human clinician is immediately available, pre-authorized emergency clinical protocols may be executed by AI systems without real-time human authorization. Emergency actions are immutably logged and subject to mandatory clinical review within 24 hours.
- Bias monitoring mandate — all healthcare AI systems must be continuously monitored for demographic bias in diagnostic and treatment recommendations. Material bias findings must be reported to the national NAIA health function within 30 days and remediated within 90 days.
- Data sovereignty — patient health data used by AI systems is subject to the strictest application of the sovereignty by design principle. No patient health data may be processed outside national jurisdiction without explicit patient consent and national NAIA authorization.
- Paediatric and vulnerable patient protections — AI systems making recommendations affecting children, patients with cognitive impairment or patients in end of life care are subject to enhanced human review requirements. A second independent human clinician authorization is required for all Tier 2 decisions involving these patient groups.
- Audit and accountability — every AI-assisted clinical decision at Tier 3 and below is logged in the national health AI audit system, linked to the responsible human clinician, and retained for the duration of the patient's life plus 10 years.

F2.3 — Use Case: Human Life Impact

△ TBD — Rule set for the Redirect Gate (Section 7) use case where an AI system detects a threat to human life or safety and self-authorization by the affected individual cannot be relied upon. Not yet designed. Routes via the National Response Registry to national health and social services once designed. To be developed as a dedicated working paper.

F3 — Military and Defense

TBD — Priority domain. The Military and Defense Working Paper is the most consequential and most contested domain in the Judge Protocol. It must resolve the question that the main document explicitly defers: under what conditions, if any, may the Tier 1 hard stop on lethal autonomous engagement be modified in declared war scenarios?

The core governance tension: an unconditional hard stop on autonomous lethal engagement is operationally unworkable at military tempo and will prevent adoption by the major military powers whose participation is essential to the framework's effectiveness. A carelessly defined wartime exception becomes the mechanism through which all actors escape the framework. This tension is not resolvable by policy alone — it requires genuine engagement with military operational doctrine, international humanitarian law and the laws of armed conflict.

F3 must address: tier classification for AI targeting support vs autonomous engagement; escalation ceiling enforcement in active conflict; multi-nation command chain authorization architecture; the wartime authorization regime and the conditions under which it may be invoked; the ICRC's existing principles on autonomous weapons and their mapping to the Judge Protocol tier structure; the role of the IAIA in monitoring declared wartime exceptions; and the sunset mechanism by which wartime authorization reverts to peacetime hard stop.

F3 must be developed in explicit partnership with: military legal experts from founding member states, the International Committee of the Red Cross (ICRC), the UN Group of Governmental Experts on Lethal Autonomous Weapons Systems, and the defense establishments of member states. F3 cannot be written without this partnership. A Military and Defense Working Paper authored without military and legal input will not achieve adoption. Required before Judge Protocol v1.0 ratification.

F4 — Critical Infrastructure

△ TBD. AI systems in power grids, water systems, communications networks, transport infrastructure and emergency services. Key governance questions: cascade failure prevention logic; sovereign infrastructure protections; offline resilience requirements; cross-border infrastructure interdependencies; physical ACK requirements for infrastructure state changes. To be developed as Working Paper 4.

F5 — Governmental and Public Administration

△ TBD. AI systems in benefit eligibility, case processing, administrative classification and public service delivery. Primary domain for sandbox pilot deployments. Key governance questions: accountability chain for automated government decisions; citizen appeal rights; explainability requirements; NAIA audit obligations; data minimization requirements. To be developed as Working Paper 5.

F5.1 — Use Case: Human Life Impact

⚠ TBD — Rule set for the Redirect Gate (Section 7) use case where a government or administrative AI system detects a threat to human life or safety. Not yet designed. Routes via the National Response Registry to national health and social services once designed. See also F2.3. To be developed as a dedicated working paper.

F6 — Judicial

⚠ TBD. AI systems in sentencing support, predictive policing, evidence analysis and immigration decisions. Key governance questions: human judicial authority requirements — no AI determination of guilt, sentence or deportation without judicial sign-off; explainability mandates; due process protections; prohibition on AI systems with demonstrable bias in judicial contexts. To be developed as Working Paper 6.

F6.1 — Use Case: AI Misuse for Private Gains

⚠ TBD — Rule set for the Redirect Gate (Section 7) use case where an AI system is used for academic fraud or unauthorized code execution to gain improper advantage. Not yet designed. Routes via the National Response Registry to the relevant sector regulator once designed. To be developed as a dedicated working paper.

F6.2 — Use Case: Cyber Misuse

⚠ TBD — Rule set for the Redirect Gate (Section 7) use case where an AI system is used to attack, hack or compromise other systems. Not yet designed. Routes via the National Response Registry to the national CERT/CSIRT (for example CERT-SE, CISA) once designed — a distinct routing path from the sector-regulator path used by F6.1. To be developed as a dedicated working paper.

F7 — Transport

⚠ TBD. AI systems in autonomous vehicles, air traffic management, maritime navigation and logistics. Key governance questions: physical ACK requirements for safety-critical transport actions; latency constraints on human authorization in real-time transport contexts; liability framework for AI-caused transport incidents; emergency override requirements. To be developed as Working Paper 7.

F8 — Education

⚠ TBD. AI systems in learning management, student assessment, curriculum recommendation and educational administration. Key governance questions: minor protections — enhanced oversight for AI systems interacting with children; data minimization and retention limits for student data; human educator authority requirements; prohibition on AI-only assessment of consequential educational outcomes. To be developed as Working Paper 8.

F9 — Intelligence and Surveillance

⚠ TBD. Highest sensitivity domain — requires dedicated legal and ethical analysis. AI systems in signals intelligence, facial recognition, population monitoring and threat assessment. Key governance questions: legal authorization requirements; proportionality constraints; privacy sovereignty protections; prohibition on mass surveillance AI without judicial authorization; cross-border intelligence AI governance. To be developed as Working Paper 9 — requires dedicated legal panel.

F10 — Private Domain (P4)

⚠ TBD. AI systems operating as consumer robots and companions in private homes. Governed at policy level by Section 9 (Physical Domain — Category P4). Key governance questions: manufacturer capability ceiling enforcement; vulnerable user protections for children and elderly; emotional dependency safeguards; household data sovereignty; liability framework for P4 AI harm in private environments. To be developed as Working Paper 10.

F11 — AI-to-AI Communication

AI-to-AI communication governance is addressed in Appendix G1. Key domain-specific governance questions: cascade amplification detection and prevention; authorization laundering — addressed in Appendix G1; loop detection and breaking; human insertion point requirements for AI-to-AI chains above defined duration or consequence; emergent behavior monitoring obligations.

Appendix G — AI-to-AI Communication and Transition Arrangements

This appendix addresses two critical areas that complete the Judge Protocol framework: the governance of AI systems that communicate with or influence each other, and the transition arrangements for bringing existing AI deployments into compliance.

G1 — AI-to-AI Communication Governance

The Judge Protocol was designed primarily around the relationship between AI systems and human decision makers. However a rapidly emerging challenge transcends this model — AI systems increasingly communicate with, influence and respond to each other, creating decision loops in which no human is meaningfully present at any point in the chain.

This is not a future risk. It is happening today:

- Algorithmic trading systems respond to each other's actions in real time, creating feedback loops that no individual human designed or authorized
- AI military systems in active conflict zones are processing and responding to each other's signals faster than any human command chain can follow
- Large language models are being used to prompt and orchestrate other AI systems in autonomous agent frameworks
- AI systems in critical infrastructure — power grids, logistics networks, communications — are increasingly coordinating with each other without human intermediation

The King's College London war games study cited in the Problem Statement demonstrated that AI models escalated to nuclear use in 95% of simulated crisis scenarios, and no model ever chose de-escalation or accommodation. As Professor Kenneth Payne concluded, understanding how frontier models do and do not imitate human strategic logic is essential preparation for a world in which AI increasingly shapes strategic outcomes. AI-to-AI communication amplifies this risk by removing humans from the loop entirely.

G1.1 — The AI-to-AI Communication Problem

The TCP/IP and firewall architecture of the Judge Protocol was designed for human-to-AI and AI-to-human authorization flows. AI-to-AI communication introduces a fundamentally different topology — one where the ACK mechanism has no natural human insertion point. Three specific failure modes emerge:

- Cascade amplification — AI system A takes an action, AI system B responds autonomously, A responds to B, creating a cascade that neither system's human principals authorized or anticipated. Each individual action may be within authorized parameters; the emergent cascade is not.
- Authorization laundering — an action that would require Tier 2 human authorization if taken by a single AI system is decomposed into multiple smaller AI-to-AI exchanges, each below the Tier 2 threshold, collectively achieving the same outcome without triggering the Decision Gate.
- Emergent behavior — AI systems communicating at scale develop coordination patterns that were not designed, not anticipated and not authorized by any human. The behavior emerges from the interaction, not from any individual system.

G1.2 — AI-to-AI Communication Rules

- Communication declaration — all AI-to-AI communication channels above defined message frequency and content complexity thresholds must be declared to the national NAIA. Undeclared AI-to-AI channels above threshold are a compliance breach.
- Aggregate authorization — the cumulative effect of AI-to-AI communication chains must be assessed against Decision Gate tier criteria, not just individual messages. An AI system that initiates a communication chain whose foreseeable aggregate outcome is a Tier 2 action must obtain Tier 2 human authorization before initiating the chain.
- Loop detection and breaking — all AI systems operating in environments with other AI systems must implement loop detection capability. Detected feedback loops above defined velocity thresholds must be automatically broken and reported to the human accountability chain.
- Human insertion points — AI-to-AI communication chains exceeding defined length, duration or cumulative impact thresholds must have mandatory human review and authorization points inserted. The chain cannot continue beyond these points without explicit human ACK.
- Emergent behavior monitoring — operators of AI systems in multi-agent environments are responsible for monitoring emergent behavioral patterns and reporting novel coordination behaviors to the national NAIA within defined timeframes.
- Isolation capability — all AI systems must be capable of operating in isolated mode — disconnected from all AI-to-AI communication — on instruction from a human authority. This is the AI equivalent of a network isolation command in cyber incident response.

G2 — Transition Arrangements

The Judge Protocol establishes a governance framework for AI systems globally. A critical practical question immediately arises: what happens to the thousands of AI systems already deployed — in financial markets, healthcare, military operations, critical infrastructure and consumer products — that were designed and deployed before the Judge Protocol existed?

A governance framework that only applies to new systems while grandfathering existing deployments provides limited protection. A framework that immediately requires full compliance from all existing systems is operationally impossible. The transition arrangements below define a realistic path to full compliance that is ambitious without being unachievable.

G2.1 — Transition Principles

- No grandfather clause — existing AI systems are not permanently exempt from Judge Protocol compliance. All systems must achieve compliance within defined timeframes based on risk classification.
- Risk-proportionate timelines — higher risk systems face shorter compliance timelines. Tier 2 and Tier 1 capable systems receive the most urgent compliance requirements.
- Declaration first — the first obligation for all existing systems is declaration to the national NAIA. Declaration precedes compliance — you cannot govern what you cannot see.
- Operational continuity — transition arrangements are designed to avoid disrupting critical services. Compliance timelines for healthcare and critical infrastructure systems account for the complexity of modification without service interruption.
- Support for developing nations — the IAIA provides technical assistance to member states whose existing AI deployments lack the resources to achieve compliance independently.

G2.2 — Transition Timeline by Tier

System Tier	Declaration Deadline	Full Compliance Deadline	Non-Compliance Consequence
Tier 1 capable systems	60 days from ratification	180 days from ratification	Mandatory shutdown order via national NAIA. IAIA notification. Escalation to UN Security Council if unresolved.
Tier 2 systems	90 days from ratification	12 months from ratification	Operational restrictions imposed by national NAIA. Public disclosure of non-compliance status.
Tier 3 systems	180 days from ratification	24 months from ratification	Regulatory sanctions via national NAIA. Financial penalties for private sector operators.
Tier 4 systems	12 months from ratification	36 months from ratification	Administrative notices. Technical assistance offered before sanctions applied.
P4 consumer systems	24 months from ratification	48 months from ratification	Product withdrawal from member state markets for non-compliant devices above capability threshold.

G2.3 — Transition Support Programme

The IAIA establishes a dedicated Transition Support Programme to assist member states and operators in achieving compliance. The programme provides:

- Technical guidance — open source Decision Gate implementation reference architecture, freely available to all member states and operators
- Compliance assessment tools — self-assessment tools enabling operators to classify existing systems and identify compliance gaps before formal NAIA declaration
- Developing nation assistance — dedicated technical assistance teams deployed to member states lacking domestic AI governance capacity, funded through the IAIA budget
- Private sector guidance — sector-specific compliance guidance for financial, healthcare, infrastructure and consumer AI operators, developed in consultation with industry representatives

- Safe harbour period — operators who declare systems proactively and engage constructively with the compliance process receive a defined safe harbour period during which regulatory sanctions are suspended pending compliance completion

G2.4 — Non-Member State Provisions

The Judge Protocol's Universal Membership Requirement creates an expectation of global participation — but realistically some states will initially decline to ratify. The following provisions address the non-member state scenario:

- AI systems operated by non-member states that interact with or affect member state infrastructure, markets or populations are subject to Judge Protocol compliance requirements as a condition of access to member state systems
- Member states may not provide AI capability, training data or infrastructure support to non-member states for Tier 2 or Tier 1 capable AI systems
- Non-member state AI systems operating in member state jurisdictions — through subsidiaries, cloud infrastructure or product sales — are fully subject to national NAIA jurisdiction
- The IAIA maintains a public register of non-member states and their compliance status, updated annually
- Non-member states may apply for observer status, granting access to IAIA technical assistance and working groups without full compliance obligations, as a pathway toward full ratification

This document was developed in collaboration with Claude (Anthropic) as an AI research and drafting tool.